

A closed-form goodness-of-fit reference for penalised logistic regression in the proportional regime

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Abstract

Grouped goodness-of-fit tests for logistic regression, such as the Hosmer–Lemeshow test, assume maximum likelihood fitting. Under ridge regression, shrinkage displaces the grouped residuals, and removing it leaves a first-order reference that is conservative. The residuals are standardised by the fitted Bernoulli variance, which shrinkage inflates towards one quarter; the resulting deflation removes 19 per cent of the statistic when predictors are a quarter of the sample, and any residual diagnostic standardised by a fitted variance inherits the error. Evaluating the variance at the debiased fit makes the reference asymptotically exact at fixed dimension. In the proportional regime of high-dimensional logistic regression we estimate the null variance from one fit by the observable adjustments of regularised M-estimation, and identify the remaining term as response-dependent grouping. The resulting test of calibration along the fitted index, CALM, needs no resampling: on a smooth basis whose degree adapts to an observable measure of index noise it matched the bootstrap in power and never exceeded its level up to aspect ratio 0.4 in every design with balanced outcomes. The decile basis needs a calibrated correction that does not transfer, and neither basis survives strongly unbalanced outcomes at those ratios. An R implementation is provided.

Keywords calibration; goodness-of-fit; high-dimensional logistic regression; Hosmer–Lemeshow test; proportional regime; ridge regression

1. Introduction

Penalised logistic regression is widely used to build a clinical prediction model when the number of candidate predictors is large relative to the number of events (Riley et al., 2021; Pavlou et al., 2024), and the fitted model is then checked for calibration in the usual way, almost always with a grouped goodness-of-fit test such as the Hosmer–Lemeshow test (Hosmer and Lemeshow, 1980; Hosmer et al., 1997). This combination is not valid. Under a ridge penalty the grouped standardised residuals acquire a non-centrality induced by shrinkage, and at the penalty that cross-validation selects the test rejects correctly specified models between 92 and 100 per cent of the time (Ebrahim, 2026a). Subtracting an estimate of the shrinkage displacement removes that non-centrality and restores the ordinary maximum likelihood covariance to first order, but the first-order reference distribution it restores is itself wrong: it is conservative, and increasingly so as the ratio of predictors to observations grows, so that until now a valid test could be had only by pre pivoting (Beran, 1987, 1988) — which Section 5 shows also loses its level at the top of the range treated here — a bootstrap in which the penalised model is refitted in every resample. The defect is in the standardisation itself, and it is not confined to grouped tests.

The explanation is simple once seen. The grouped residuals are standardised by the Bernoulli variance $\hat{\pi}_i(1 - \hat{\pi}_i)$ evaluated at the *fitted* probabilities. Ridge regression pulls every fitted probability towards one half, where $\pi(1 - \pi)$

is largest, so the standardisation uses a variance that is too large and every residual looks smaller than it is. The error is of order λ/n , where λ is the penalty and n the sample size; it is negligible in the asymptotic regime of the first-order theory and material at the penalties analysts actually use — 7 per cent of the statistic at $\lambda/n = 0.2$ and 19 per cent when the number of predictors is a quarter of the sample size. Evaluating the variance at the debiased fit instead makes the analytic reference asymptotically exact wherever the first-order theory applies. The point is not confined to grouped tests. Every residual diagnostic for a penalised generalised linear model that standardises a residual by its fitted variance divides by a quantity that shrinkage has moved towards the maximum of the variance function, and so reports a statistic that is systematically too small; Pearson-type and deviance-type statistics for penalised fits should show the same error.

In the proportional regime, where $p/n \rightarrow \kappa \in (0, 1)$, two further things happen, and both are dealt with here. First, the debiased fit is itself noisy, so its Bernoulli variance understates the truth; we use the observable adjustments of Bellec (2025) for regularised M-estimation to estimate the null variance from the fit alone. Second, grouping on a noisy fitted index selects observations by their own noise, which leaves a non-centrality in the corrected residual. We show that this term is of order κ within a design family, that under balanced outcomes it is concentrated almost entirely in the cubic direction of the group index, and that it is then removed by a basis whose polynomial degree is chosen from an observable measure of index noise — a rule that the attenuation law of Ebrahim (2026a) already recommends on grounds of power. The result is a test with a closed-form reference distribution and no resampling. We call it CALM, for calibration assessment under lambda-shrunk models; the name records what the statistic is computed from and where, and Section 2.2 states precisely which hypothesis it tests, which is not the calibration of the shrunk probabilities themselves. One conclusion deserves to be stated at the outset, because it decides the design: in the proportional regime the choice of basis is not a matter of taste but of validity. A basis that spends every group direction cannot avoid the direction the selection term occupies, and can only absorb it into the reference through a constant that must be calibrated; a basis of low polynomial degree can leave that direction out, and then no constant is needed. The EDGE basis of Ebrahim and El-Kotory (2026), introduced there to direct a test at the departures a calibration curve can show, turns out to be the ingredient that makes an untuned closed-form reference possible here. Tests for high-dimensional regression built by other routes — the residual prediction tests of Shah and Bühlmann (2018) for linear models and Janková et al. (2020) for generalised linear models, and the coefficient tests of Guo and Chen (2016) — ask whether the model is correct as a function of the covariates, or whether a block of coefficients is zero; the statistics treated here ask whether the deployed predictor is calibrated along its own index, and Section 5.6 shows that the two families reject on essentially disjoint datasets: against an omitted interaction off the index the residual prediction test rejects 0.877 at fixed p where CALM rejects 0.058, and against a departure along the index the ordering reverses. The present paper supplies the analytic reference that the second family lacked.

It is worth saying at the outset what a test of this kind can and cannot deliver. The ideal is validity at any aspect ratio together with power against every departure from the logistic form, and that ideal is out of reach for any test built on the fitted index. Grouping on a noisy index attenuates a departure of degree k by ρ^{2k} , where ρ is the correlation between the fitted and the true linear predictor (Proposition 2), and ρ is about 0.7 at $\kappa = 0.25$, so a cubic departure arrives with about a tenth of its non-centrality and no grouped test on that index can find it, whatever reference it uses. What is attainable, and what this paper delivers, is a valid reference computed from a single fit together with power against the departures the index preserves. Section 5.8 shows that a bootstrap with 399 refits reaches no further.

The contributions are as follows. (a) Theorem 1 identifies the fitted-variance deflation and gives the corrected first-order law. (b) Proposition 1 gives an observable estimate of the null variance in the proportional regime, and Section 4.3 characterises the selection non-centrality, its order in κ and its cubic structure. (c) Corollary 1 turns the attenuation law of Ebrahim (2026a), restated here as Proposition 2, into an adaptive basis that retains the directions in which a departure survives grouping and, as a consequence, removes the selection term. (d) Section 5 separates the two bases: on the adaptive EDGE basis, which by construction avoids the direction the selection term occupies, CALM held or fell below its level in all twenty-five balanced designs of Tables 2 and 3, for κ up to 0.4 and for three predictor laws, while on the decile basis it needs the calibrated correction; we report the designs in which that correction is too small, and the unbalanced designs in which neither basis holds its level. (e) Section 5.6 compares CALM with pre pivoting, with the high-dimensional test of Janková et al. (2020) and with the calibration checks in clinical use, which do not hold their level on a penalised fit.

Three kinds of statement follow, and we keep them apart throughout. Theorem 1 is proved here, and Corollary 1 follows from the attenuation law of Ebrahim (2026a), restated as Proposition 2. Proposition 1 rests on the theorems of Bellec (2025) under a Gaussian design and on the consistency of a one-dimensional scale estimate, which we assume. The relation of Section 4.3 that supplies the inflation factor is fitted on eight simulation cells and is not derived: Section 5.4 reports its behaviour in sixteen cells that did not calibrate it, including five in which the decile statistic fails. The statistic we recommend, on the adaptive basis of Corollary 1, does not depend on that constant wherever the rule takes degree two, which it did in every proportional-regime cell we examined: the inflation it applies is then inert, at most 1.05. Where the index is accurate enough for the rule to keep the cubic, as on the data of Section 6, the constant returns.

Section 2 sets out the shrinkage-corrected statistics and the first-order law we take as given, Section 3 proves the fitted-variance deflation, Section 4 builds the observable reference in the proportional regime and defines CALM, Section 5 reports the numerical study, Section 6 returns to the glaucoma data of Ebrahim (2026a), and Section 7 states what is proved, what is calibrated and where the boundary lies. The proofs are short and are given where the results are stated; the behaviour of the correction under penalties other than the ridge is reported in the online supplementary material.

2. The shrinkage-corrected statistics and their first-order law

2.1. Notation

Let $y_i \in \{0, 1\}$ and $\mathbf{x}_i \in \mathbb{R}^p$, $i = 1, \dots, n$, and write X for the $n \times (p + 1)$ design matrix including the intercept column. The logistic model specifies $\pi_i(\beta) = \{1 + \exp(-\mathbf{x}_i^\top \beta)\}^{-1}$, and penalised estimation solves $\hat{\beta} = \arg \max_{\beta} \{\ell(\beta) - \frac{1}{2} \lambda \beta^\top D \beta\}$ with $D = \text{diag}(0, 1, \dots, 1)$, the intercept unpenalised (le Cessie and van Houwelingen, 1992). The penalty is on the scale of the log-likelihood; the implementation of Friedman et al. (2010) minimises $-n^{-1} \ell(\beta) + \frac{1}{2} \lambda_g \|\beta_{-0}\|^2$, so $\lambda = n \lambda_g$, and we report both scales. The correspondence assumes the penalty is applied on the scale of the design as supplied. That implementation standardises the columns by default and returns coefficients on the original scale, on which its penalty is the generalised ridge $\frac{1}{2} \lambda_g \sum_j s_j^2 \beta_j^2$ with s_j the standard deviation of the j th predictor; an analyst should therefore either standardise the design before fitting, as we do throughout, or use the generalised-ridge form of Section 5.9. Write $\hat{\pi}_i = \pi_i(\hat{\beta})$, $W = \text{diag}\{\hat{\pi}_i(1 - \hat{\pi}_i)\}$, $F = X^\top W X$, $K = \lambda D$ and $M = F + K$. Let C be the $G \times n$ indicator matrix of the G equal-frequency groups of the fitted probabilities, $V = C W C^\top$, and

$$\mathbf{r} = V^{-1/2} C(\mathbf{y} - \hat{\pi}), \quad U = V^{-1/2} C W X. \quad (1)$$

The Hosmer–Lemeshow statistic is $\|\mathbf{r}\|^2$ up to the choice of denominator. Two bases are used throughout: the *decile* basis spends all G directions, and the *EDGE* basis of Ebrahim and El-Kotory (2026), which we introduced there for unpenalised fits, projects $\mathbf{r}$ onto the orthogonal polynomials of degree one to k in the group-mean fitted probability, Z being the resulting $G \times k$ matrix, with $k = 3$ as the default. Smoothing the residuals against the fitted probability is the older route to the same idea (le Cessie and van Houwelingen, 1991); the EDGE basis differs in fixing a low-dimensional polynomial space, which is what makes the degree rule of Corollary 1 possible. We write $P_Z = Z(Z^\top Z)^{-1} Z^\top$.

2.2. The corrected statistics

Under a penalty, $\mathbb{E}(\mathbf{r}) \neq \mathbf{0}$ even when the model is correct, because the fitted probabilities are shrunk; the correction (2) below removes that mean. Ebrahim (2026a) derived the displacement $\boldsymbol{\mu}_K = U M^{-1} K \beta$ to first order and showed that with the one-step debiased estimate $\tilde{\beta} = \hat{\beta} + F^{-1} K \hat{\beta}$ and the plug-in $\hat{\boldsymbol{\mu}} = U M^{-1} K \hat{\beta}$, the corrected residual

$$\mathbf{v} = \mathbf{r} - \hat{\boldsymbol{\mu}} \quad (2)$$

satisfies, to first order and under regularity conditions with $\lambda = o(n)$ and p fixed,

$$\mathbf{v} = A^*(\mathbf{y} - \boldsymbol{\pi}) + o_p(1), \quad A^* = V^{-1/2} C - U F^{-1} X^\top, \quad (3)$$

where $\pi = \pi(\beta)$ is the true probability vector. The shrinkage-corrected statistics are

$$\text{SC.HL} = \|v\|^2, \quad \text{SC.EDGE} = v^\top P_Z v. \quad (4)$$

Equation (3) is Proposition 2 of Ebrahim (2026a), proved there under Assumption 1 below, which we restate so that this paper is self-contained. We use it as given; the argument of the present paper begins where that one stops, with the covariance of the right-hand side.

Assumption 1 The true probabilities satisfy $\pi_i(\beta) \in [\epsilon, 1 - \epsilon]$ for some $\epsilon > 0$ uniformly in i ; $n^{-1}F \rightarrow \Sigma$ positive definite, and λ is non-random with $\lambda = o(n)$; the number of groups G is fixed and each group receives a fixed positive proportion of the observations; $\max_{i \leq n} \|x_i\| = O_p(1)$; and p is fixed as $n \rightarrow \infty$.

The last two conditions are the ones that matter for what follows. The fixed grouping is an assumption rather than a consequence, as Ebrahim (2026a) notes; the second is the subject of the following remark.

Remark 1 Equation (3) is proved for p fixed and $\lambda = o(n)$. Sections 4.2 to 4.5 apply it in the proportional regime of Section 4, where p/n tends to a positive constant and the penalties used have λ/n of order one, so its validity there is assumed and not proved. Closing that gap would require an expansion of the grouped residual in the proportional regime, which we do not attempt here; what can be done is to check the consequence, and Section 5 does so in twenty-five designs. Every statement in Sections 4.2 to 4.5 should be read subject to this remark, and Section 7 returns to it.

The null hypothesis must be stated with care, because two hypotheses that coincide under maximum likelihood separate under a penalty (Ebrahim, 2026a). The structural null H_0^{str} says that the logistic form is correct: $\mathbb{E}(y_i | x_i) = \pi_i(\beta)$ for some β . The calibration null H_0^{cal} says that the deployed probabilities are calibrated: $\mathbb{E}(y_i | \hat{\pi}_i) = \hat{\pi}_i$. Under a penalty a correct model delivers shrunk probabilities, so H_0^{str} can hold while H_0^{cal} fails by an amount the analyst chose with λ (van Houwelingen and le Cessie, 1990). The statistics of (4), and CALM, test H_0^{str} along the fitted index: a non-rejection says the form is right along that index, not that the shrunk probabilities can be used without recalibration.

2.3. Exact tail probabilities

If $v \sim \mathcal{N}(0, \Sigma)$ then $\|v\|^2$ is distributed as $\sum_l \omega_l \chi_{1,l}^2$ with ω_l the eigenvalues of Σ , and $v^\top P_Z v$ as $\sum_l \nu_l \chi_{1,l}^2$ with ν_l the eigenvalues of $\Sigma^{1/2} P_Z \Sigma^{1/2}$, of which at most k are non-zero. Tail probabilities of such weighted sums are available to any accuracy by the algorithm of Davies (1980), with Imhof (1961) as a fallback; we use the implementation of Duchesne and Lafaye de Micheaux (2010). No chi-squared approximation with a nominal number of degrees of freedom is used anywhere in this paper, and nothing that follows depends on one.

3. The fitted-variance deflation

3.1. The corrected covariance

The first-order theory refers v to $\mathcal{N}(0, \Omega_{\text{MLE}})$ with $\Omega_{\text{MLE}} = I_G - UF^{-1}U^\top$, the ordinary maximum likelihood covariance of the grouped residuals, and this reference was found to be conservative in the proportional regime (Ebrahim, 2026a, Table S1). The following result says exactly what is wrong with it.

Theorem 1 Let $W_0 = \text{diag}\{\pi_i(1 - \pi_i)\}$ be the Bernoulli variance at the true probabilities, and A^* as in (3). Under the conditions of (3),

$$\text{var}(v) = A^* W_0 A^{*\top} + o_p(1). \quad (5)$$

Moreover the fitted-variance identity

$$A^* W A^{*\top} = I_G - UF^{-1}U^\top = \Omega_{\text{MLE}}$$

holds exactly for the fitted W , and

$$\|A^*W_0A^{*\top} - \Omega_{\text{MLE}}\| = O_p(\|\hat{\beta} - \beta\|) = O_p(\lambda/n + n^{-1/2}). \quad (6)$$

Proof Two conventions precede the second moments. The grouping is held as given, which Assumption 1 permits, and A^* is held at A_0^* , its value at β with that grouping; $\|A^* - A_0^*\| = O_p(\|\hat{\beta} - \beta\|)$ by the Lipschitz argument used for (6) below, so restoring the fitted A^* costs nothing at that order. And $\text{var}(\mathbf{v})$ is read as the second moment of the leading term of (3), the $o_p(1)$ remainder being negligible there provided $\{\|\mathbf{v}\|^2\}$ is uniformly integrable, which we assume rather than show. With these conventions $\text{var}(\mathbf{v}) = A_0^* \text{var}(\mathbf{y} - \pi)A_0^{*\top} + o_p(1)$, and $\text{var}(y_i - \pi_i) = \pi_i(1 - \pi_i)$ because the y_i are independent Bernoulli variables with the true probabilities; this is (5). For the identity, expand $A^*WA^{*\top}$ using the three relations $V^{-1/2}CWC^\top V^{-1/2} = I_G$, $V^{-1/2}CWX = U$ and $X^\top WX = F$, all of which hold because V , U and F are built from the same W :

$$A^*WA^{*\top} = I_G - UF^{-1}U^\top - UF^{-1}U^\top + UF^{-1}FF^{-1}U^\top = I_G - UF^{-1}U^\top.$$

For (6), $A^*(W_0 - W)A^{*\top}$ is bounded by $\|A^*\|^2 \max_i |\pi_i(1 - \pi_i) - \hat{\pi}_i(1 - \hat{\pi}_i)|$; the map $\pi \mapsto \pi(1 - \pi)$ is Lipschitz with constant one, $|\hat{\pi}_i - \pi_i| \leq \frac{1}{4}|\mathbf{x}_i^\top(\hat{\beta} - \beta)|$, and $\|\hat{\beta} - \beta\| = O_p(\lambda/n + n^{-1/2})$ by the expansion in Proposition 1 of Ebrahim (2026a), the λ/n term being the shrinkage displacement $M^{-1}K\beta$ and the $n^{-1/2}$ term the estimation error. $\square$

The point of the theorem is the direction of the error, which we call the *fitted-variance deflation*. The first-order reference is not wrong in its algebra: Ω_{MLE} is exactly $A^*WA^{*\top}$. The error is in W . Ridge regression pulls the fitted linear predictor towards zero and $\pi(1 - \pi)$ is maximised at one half, so $\hat{\pi}_i(1 - \hat{\pi}_i) \geq \pi_i(1 - \pi_i)$ for every observation whose fitted linear predictor is smaller in absolute value than the true one; this need not hold for every observation, but it holds for the great majority under a penalty towards zero, the standardisation in (1) then divides by too large a variance, and $\|\mathbf{v}\|^2$ is systematically smaller than $\text{tr}(\Omega_{\text{MLE}})$.

The identity is worth stating on its own. The maximum likelihood covariance of the grouped residuals, on which the classical grouped tests rest, is not an approximation waiting to be improved: it is exactly the covariance of the corrected residual with the Bernoulli variance evaluated at the fitted probabilities. Under maximum likelihood the fitted variance is the right variance and the reference is right; under any penalty it is the wrong variance, and the theorem says by how much. The proof uses only that V , U and F are built from the same W , so the same statement holds for any residual diagnostic whose covariance is assembled from the fitted weights. The correction is of a different kind from that of Osius and Rojek (1992), which adjusts the null variance of the Pearson statistic for the estimation of β under maximum likelihood; here the error is in the point at which the variance is evaluated, not in ignoring estimation.

Write

$$\delta = \mathbb{E}(\text{SC.HL}) / \text{tr}(\Omega_{\text{MLE}}) \quad (7)$$

for the deflation ratio, the mean of the statistic divided by the mean its reference implies for it. A reference that is right has $\delta = 1$; Theorem 1 says that under a penalty $\delta < 1$, and by how much. We estimate it by the Monte Carlo mean under a correct model and report it for every design in the paper, because it is the single number that says whether a residual diagnostic for a penalised fit has been referred to the right law, and any simulation study of such a diagnostic can compute it. At fixed $p = 5$ with $\lambda/n = 0.2$, the cell the theorem covers, $\delta = 0.93$. In design B of Section 5, at $\kappa = 0.25$ and $\lambda = 416$, the observed mean of SC.HL under a correct model is 4.87 against $\text{tr}(\Omega_{\text{MLE}}) = 6.02$, so $\delta = 0.81$, a deficit of 19 per cent; that second figure is measured at an aspect ratio the theorem does not reach, and we report it because it is the size of the error in the regime the rest of the paper treats. Figure 1 shows δ in all eight cells examined. The bootstrap of Ebrahim (2026a) was never affected, because it generates from the debiased fit and so carries the correct variance implicitly; the analytic reference was.

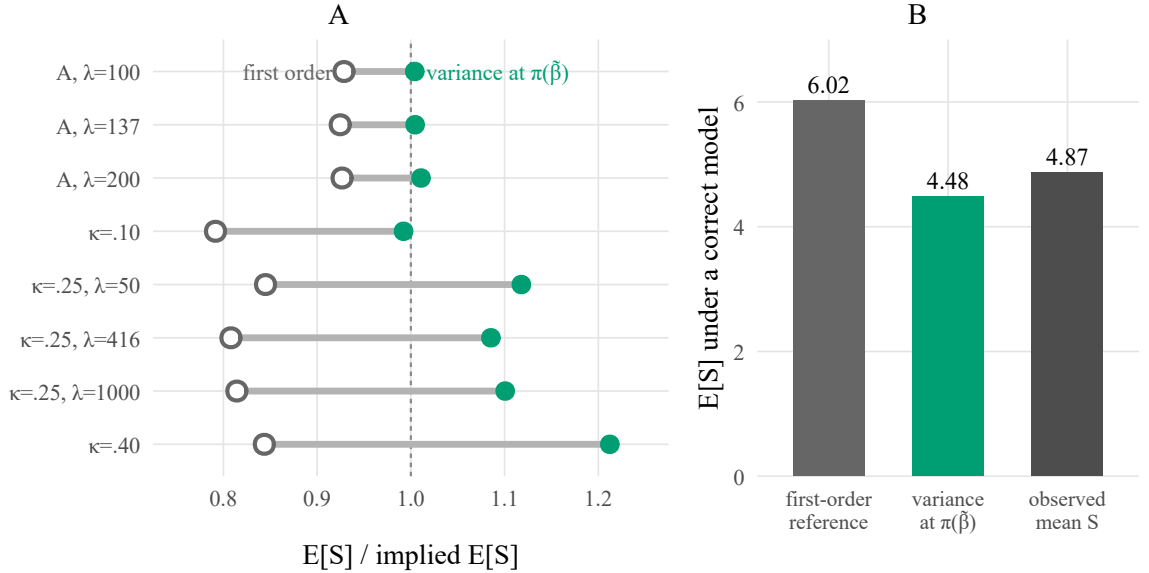


Figure 1 The fitted-variance deflation. A: the deflation ratio δ of (7) falls short of one in all eight cells under the first-order reference (open circles), and the shortfall is removed by evaluating the variance at the debiased fit (filled circles). B: the three quantities at $\kappa = 0.25$, $\lambda = 416$.

3.2. The variance at the debiased fit

Theorem 1 suggests the obvious repair: replace W by an estimate of W_0 . The natural one is $\widetilde{W} = \text{diag}\{\tilde{\pi}_i(1 - \tilde{\pi}_i)\}$ with $\tilde{\pi}_i = \pi_i(\hat{\beta})$, the Bernoulli variance at the debiased fit, and the reference

$$\Sigma_d = A^* \widetilde{W} A^{*\top}. \quad (8)$$

This is exactly the choice the pre pivoting bootstrap makes implicitly, since it generates resampled responses from $\pi(\hat{\beta})$. Table 2 reports the consequence. At fixed p δ moves from 0.93 to 1.00–1.01 and the size from 0.02–0.03 to 0.04–0.05; at $\kappa = 0.10$ from 0.79 to 0.99 and 0.006 to 0.054. Wherever the first-order theory applies, replacing the variance is the whole correction.

At $\kappa \geq 0.25$, however, (8) over-corrects: δ rises above one and the size to 0.07–0.13, most where the penalty is lightest. The reason is that $\hat{\beta}$ is now a noisy estimate — at $\kappa = 0.25$ its correlation with the true index is about 0.7 — so $\tilde{\pi}_i$ is *more* extreme than π_i and $\widetilde{W}$ understates the null variance. Two things are needed in the proportional regime: a variance estimate that does not inherit the noise of $\hat{\beta}$, and an account of a second term that the proportional regime introduces. The next section supplies both.

4. An observable reference in the proportional regime

4.1. Observable adjustments

Bellec (2025) studies regularised M-estimators $\hat{\beta} = \arg \min_b n^{-1} \sum_i \ell_{y_i}(x_i^\top b) + g(b)$ in single-index models $y_i = \mathcal{F}(x_i^\top w, \varepsilon_i)$ with Gaussian design, in the regime where p/n is bounded, and shows that the mean-field description of $\hat{\beta}$ and of the fitted values $X\hat{\beta}$ obtained by Sur and Candès (2019), Salehi et al. (2019) and others can be expressed through quantities computed from the data alone. We need three of them, written in our notation. Let $\psi = y - \hat{\pi}$ be the raw residual, let $\text{df} = \text{tr}(XM^{-1}X^\top W)$, let $\Gamma = W - WXM^{-1}X^\top W$ and $\hat{\gamma} = \text{df} / \text{tr}(\Gamma)$; and let $\hat{a}^2$ and $\hat{\sigma}^2$ be the estimators of Bellec’s equation (3.20) of, respectively, the squared inner product of $\hat{\beta}$ with the index direction and the squared norm of its component orthogonal to the index. Then Theorem 4.3 of Bellec (2025) gives, for the

logistic loss and each observation,

$$\mathbf{x}_i^\top \hat{\boldsymbol{\beta}} - \hat{\gamma} \psi_i \approx a_* U_i + \sigma_* Z_i, \quad (9)$$

where $U_i = \mathbf{x}_i^\top \mathbf{w}$ is the true index standardised to unit variance, Z_i is standard normal and independent of U_i , and (a_*, σ_*) are consistently estimated by $(\hat{a}, \hat{\sigma})$ (his Theorem 4.4). Equation (9) says that an observable quantity, $m_i = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}} - \hat{\gamma} \psi_i$, equals the true index up to scale plus independent Gaussian noise of known variance. Two points of implementation must be stated. Bellec's model has no intercept and his penalty is strongly convex in every direction, while ours leaves the intercept unpenalised; we centre the design and the fitted index before applying (9) and refit the intercept in (11) below. And his estimator of $\hat{a}^2$ involves $\Sigma_x^{-1/2}$ with Σ_x known; we replace Σ_x by the sample covariance of the centred design, a plug-in whose error at p/n up to 0.4 is not covered by his Theorem 4.4. Its size can be measured, since in a simulation the true Σ_x is known: over 200 replications the plug-in raises $\hat{\rho}$ from the oracle value by 0.030 at $\kappa = 0.25$, 0.042 at $\kappa = 0.40$ and 0.034 in the design with weak covariate correlation. The bias is small but it is one-sided, and it is in the direction that makes the rule of Corollary 1 keep the cubic direction slightly too often, which the threshold τ has to absorb.

4.2. The de-noised null variance

Given (9) and a Gaussian design, the conditional law of the true index given the fit is Gaussian:

$$a_* U_i \mid m_i \sim \mathcal{N}\left(\frac{\hat{a}^2}{\hat{a}^2 + \hat{\sigma}^2} m_i, \frac{\hat{a}^2 \hat{\sigma}^2}{\hat{a}^2 + \hat{\sigma}^2}\right). \quad (10)$$

The true linear predictor is $\eta_{0i} = \alpha + c a_* U_i$ for an intercept α and a scale c that the single-index model leaves free but the known logistic link identifies. We estimate (α, c) by maximising the marginal likelihood of $\mathbf{y}$ given the conditional law (10), a one-dimensional integral per observation evaluated by twenty-point Gauss–Hermite quadrature, and define

$$\hat{w}_{0i} = \mathbb{E}\{\text{expit}(\hat{\alpha} + \hat{c}\zeta) [1 - \text{expit}(\hat{\alpha} + \hat{c}\zeta)] \mid m_i\}, \quad \text{where } \zeta \text{ has the conditional law (10),} \quad (11)$$

the de-noised estimate of the null Bernoulli variance, with $\widehat{W}_0 = \text{diag}(\hat{w}_{0i})$ and the reference $\Sigma_{\text{dn}} = A^* \widehat{W}_0 A^{*\top}$.

Assumption 2 (a) The design has independent rows $\mathbf{x}_i \sim \mathcal{N}(\mathbf{0}, \Sigma_x)$ with the condition number of Σ_x bounded. (b) p/n is bounded above and, for the rates quoted from Bellec (2025), bounded away from zero, the fixed-dimension case being covered instead by Theorem 1. (c) The penalty satisfies Assumptions A and B of Bellec (2025) after the intercept is removed by centring. (d) The rows of A^* have entries of order $n^{-1/2}$, as they do when each group receives a fixed positive proportion of the observations, as in Assumption 1. (e) The logistic scale c is identified and its maximum-likelihood estimate consistent.

Conditions (a) to (c) are those of Bellec (2025), and the ridge penalty of Section 2.1 meets (c) once the intercept is removed by centring, the remaining penalty being strongly convex. Condition (d) is the proportional-regime counterpart of the fixed grouping of Assumption 1. Condition (e) is the one that is assumed rather than shown, and the discussion after the proof reports where the estimate it concerns is good and where it is not. Assumption 2 binds under the null only: it enters the reference distribution and therefore governs the size of the test, while under a departure it affects nothing but the degree that the rule of Corollary 1 selects, which is why that rule carries the two guards of Section 4.4.

Proposition 1 *Under Assumption 2 and a correctly specified logistic model, $n^{-1} \sum_i |\hat{w}_{0i} - \mathbb{E}\{\pi_i(1 - \pi_i) \mid m_i\}| \rightarrow 0$ in probability, and hence $\Sigma_{\text{dn}} - A^* \widehat{W}_0 A^{*\top} \rightarrow 0$, where $\widehat{W}_0 = \text{diag}[\mathbb{E}\{\pi_i(1 - \pi_i) \mid m_i\}]$ is the conditional null variance given the fit.*

Proof By Theorems 4.3 and 4.4 of Bellec (2025), $(\hat{\gamma}, \hat{a}^2, \hat{\sigma}^2)$ converge to $(\gamma_*, a_*^2, \sigma_*^2)$ and the representation (9) holds with an averaged Wasserstein error of order $n^{-1/2}$. With a Gaussian design, (10) is the exact conditional law of $a_* U_i$ given m_i under the limiting representation, so the right-hand side of (11), a bounded Lipschitz functional of $(m_i, \hat{a}, \hat{\sigma}, \hat{\alpha}, \hat{c})$, converges to $\mathbb{E}\{\pi_i(1 - \pi_i) \mid m_i\}$ by the continuous-mapping theorem once $(\hat{\alpha}, \hat{c})$

are consistent, which is the last part of the assumption. The statement about Σ_{dn} follows from the row-norm condition in Assumption 2: writing $\mathbf{a}_g$ for the g th row of A^* and $d_i = \hat{w}_{0i} - \mathbb{E}\{\pi_i(1 - \pi_i) \mid m_i\}$, the (g, h) entry of $A^* \text{diag}(d) A^{*\top}$ is bounded by $\max_g \|\mathbf{a}_g\|_\infty^2 \sum_i |d_i|$, which is $O(n^{-1} \sum_i |d_i|)$ by condition (d) of Assumption 2, and the first part of the proposition bounds $n^{-1} \sum_i |d_i|$. $\square$

Three remarks on the proposition. It is stated for the conditional variance given the fit, and its target is that conditional variance under the limiting representation (9) rather than the exact finite-sample conditional variance; the two differ by the Wasserstein error of Bellec's theorem. And its last hypothesis is not proved here: the scale c enters only through the known link, and consistency of its one-dimensional maximum-likelihood estimate is what one expects but is stated as an assumption so that the reader knows where the proof stops. Estimating the signal strength of a high-dimensional logistic model from data alone is a solved problem for the maximum likelihood estimator, by the ProbeFrontier method of Sur and Candès (2019) and by SLOE (Yadlowsky et al., 2021); those estimators are built for the maximum likelihood fit and do not apply to a ridge fit at a penalty of the analyst's choosing, which is why the scale is estimated here from the conditional law (10). The estimate is constrained to $c \in [0.05, 20]$, a box introduced after an unconstrained search diverged on a small number of replications. Finally, the covariance Theorem 1 calls for $A^* W_0 A^{*\top}$, while the proposition delivers $A^* \bar{W}_0 A^{*\top}$. Their difference has (g, h) entry $\sum_i a_{gi} a_{hi} d_i$ with $d_i = \pi_i(1 - \pi_i) - \mathbb{E}\{\pi_i(1 - \pi_i) \mid m_i\}$ conditionally centred and bounded by $1/4$, and weights of order n^{-1} by condition (d); the sum is $o_p(1)$ provided the d_i are weakly dependent, which we assume. Empirically the estimate is good where the penalty is moderate and poor where it is light. Across the twenty-five designs of Section 5, whose proportional-regime cells are design B' and so have a true standard deviation of the linear predictor of 2.31, it recovers that standard deviation within 3 per cent at fixed p and within 16 per cent in seventeen of the eighteen proportional-regime cells at $\lambda \geq 416$, under Gaussian, heavy-tailed and binary predictors alike, the exception being the heaviest penalty at the largest aspect ratio, $\kappa = 0.40$, $\lambda = 1000$, where it understates the scale by 24 per cent; at the lightest penalties it overstates the scale badly, by 21 per cent at $\kappa = 0.25$, $\lambda = 50$ and by 58 per cent at $\kappa = 0.40$, $\lambda = 50$. Those are also the cells in which the decile statistic is least reliable, and the two facts are connected: an overstated scale understates the null variance.

Table 2 shows what the de-noised variance buys. It coincides with (8) at fixed p , repairs the cell in which $\tilde{\beta}$ overshoots most ($\kappa = 0.25$, $\lambda = 50$: size 0.046 against 0.086), and halves the excess at $\kappa = 0.40$. A residual of one to three percentage points remains at $\kappa \geq 0.25$, and it is not a variance.

4.3. The selection non-centrality

The remaining term is a mean. Under a correct model the corrected residual should average zero in every group; at $\kappa = 0.25$ it does not. Over 300 replications of design B at $\lambda = 416$ the squared norm of the mean of $\mathbf{v}$ is 0.288 (300 replications; the Monte Carlo floor, the value expected from sampling error alone, is $\text{tr}(\Omega_{\text{MLE}})/300 \approx 0.020$), and its shape across the ten groups is the S-curve of Figure 2A. Holding the grouping fixed — forming the deciles from the fitted index of the noise-free response $\pi(\beta)$, a function of X alone — reduces it to 0.077, with the deflation ratio unchanged. Most of the term is therefore created by grouping on the fitted index. The remainder is not: 0.077 is about six times the Monte Carlo floor, and it is the second-order part of the shrinkage displacement, which the plug-in $\hat{\mu}$ removes only to first order. A delta-method expansion of $\hat{\mu}$ about $\tilde{\beta}$ predicts 0.067 for it, close to what is observed, and its effect on the size of the test is at most 0.007 in this cell.

The mechanism is regression to the mean, and we call it response-dependent grouping. The fitted index is a noisy version of the true one, and an observation reaches the top decile partly because its own noise was positive; noise is positive when y_i exceeded its expectation, so the top decile is enriched with events beyond what its probabilities justify, and the bottom decile depleted. That argument gives the existence and the order of the term in the raw grouped residual; in the corrected residual $\hat{\mu}$ absorbs most of it, and the cubic shape of what survives is an empirical finding rather than a consequence of the argument. At fixed p the self-influence of y_i on $\hat{\eta}_i$ (Pregibon, 1981) is $h_{ii} = w_i \mathbf{x}_i^\top M^{-1} \mathbf{x}_i$, of order p/n , and the effect is negligible; in the proportional regime it is of order κ . This is the random-cells effect of Moore and Spruill (1975), which is $o(1)$ in their setting and $O(1)$ here. The groups are formed in part from the responses they are then used to test.

Empirical relation 1 Let $\bar{h} = \hat{\text{df}}/n$ be the mean self-influence. Across the eight cells of Section 5, the fraction $\hat{g} = \|\mathbb{E}\mathbf{v}\|^2 / \text{tr}(\Omega_{\text{MLE}})$ satisfies $\hat{g} \approx 0.24\kappa$ (coefficient of determination 0.93), and equivalently $\hat{g} \approx 0.28\bar{h}^{1/2}$;

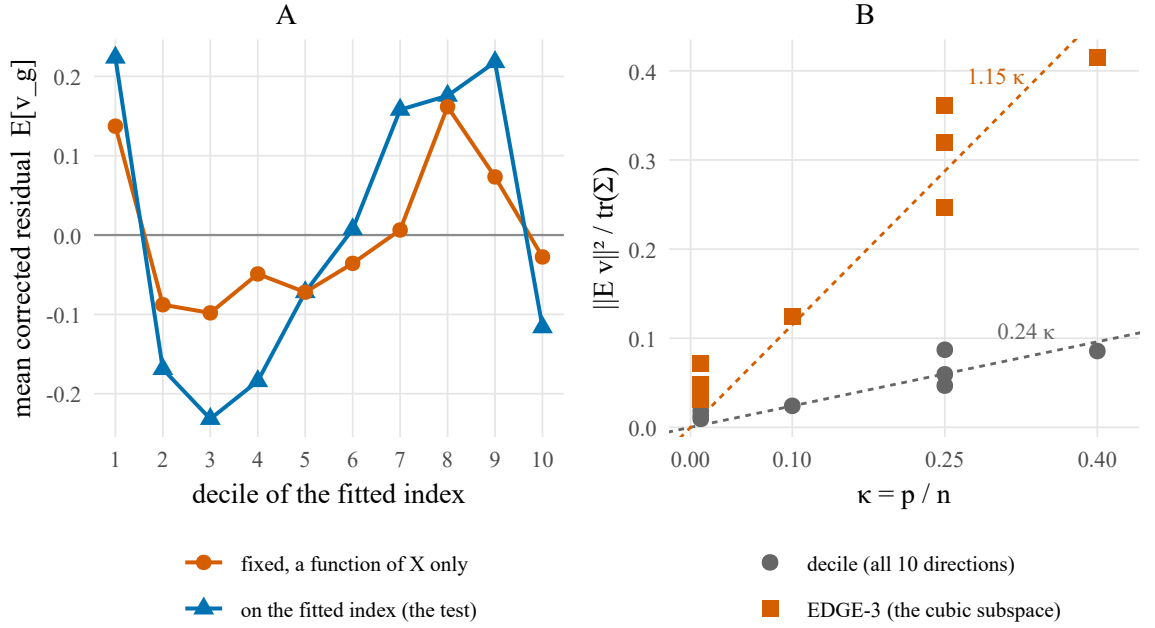


Figure 2 The selection non-centrality. A: mean corrected residual by decile at $\kappa = 0.25$, $\lambda = 416$, when the groups are formed on the fitted index and when they are held fixed as a function of the design. B: the non-centrality fraction against κ for the decile basis and for the three-dimensional EDGE subspace, with the fitted lines 0.24κ and 1.15κ .

restricted to the three-dimensional EDGE subspace, and normalised there by the trace the reference itself supplies on that subspace, $\hat{g}_E = \|P_Z \mathbb{E} v\|^2 / \text{tr}(P_Z \Sigma_{\text{dn}}) \approx 1.15 \kappa$ (0.97), and 90 to 98 per cent of $\|\mathbb{E} v\|^2$ lies in that subspace when $\kappa > 0$, almost none of it in the directions of degree at most two. The constants are fitted on those eight cells and are not derived; the relation is written in κ because κ orders the cells examined here, and Section 5.4 shows that the quantity the term really tracks is the self-influence measured against the spread of the fitted index, so the relation should not be carried to design families in which those two move apart. The two fractions are normalised differently: the decile constant is fitted against the first-order trace $\text{tr}(\Omega_{\text{MLE}})$, which exceeds $\text{tr}(\Sigma_{\text{dn}})$ across these cells, and fitted against $\text{tr}(\Sigma_{\text{dn}})$ instead it would be 0.30κ (0.96). CALM applies 0.24κ , so on the decile basis the inflation is a calibrated factor rather than an exact mean match.

Section 5.4 reports what happens in cells that were not used to fit the constants. The content of the relation is in Figure 2B. Two consequences are immediate. First, the non-centrality can be placed in the distribution by inflating the scale of the reference to $(1 + \hat{g})\Sigma_{\text{dn}}$, and because a scaled central weighted chi-squared with the correct mean has a heavier upper tail than the true non-central one, an inflation that matches the mean can only err towards conservatism. On the EDGE subspace it does match, since $\hat{g}_E$ is normalised by the same $\text{tr}(P_Z \Sigma_{\text{dn}})$ that the reference supplies there. On the decile basis it does not: $\hat{g}$ is normalised by the larger first-order trace, so the mean is under-matched and the level that CALM-D holds in Table 2 is measured rather than implied. Second, and more usefully, the term is a cubic. It lives in exactly the direction of the EDGE basis that the attenuation law says carries no power when the index is noisy.

4.4. The adaptive basis

Proposition 2 (Attenuation law, Ebrahim, 2026a) *Let η^* and $\hat{\eta}^*$ be the true and fitted linear predictors, standardised, jointly Gaussian with correlation ρ . A departure equal to the degree- k Hermite component of the*

true predictor survives grouping on the fitted predictor with amplitude ρ^k , and its non-centrality is attenuated by ρ^{2k} .

The correlation ρ is not a nuisance one must guess. Under (9) it is

$$\hat{\rho} = \frac{\hat{a}}{\sqrt{\hat{a}^2 + \hat{\sigma}^2}}, \quad (12)$$

an observable, which we call the index correlation: it is computed from one fit and it measures how much of the true index survives in the fitted one, so it is the quantity that decides what a grouped test can see. By Proposition 2 a degree- k departure reaches the groups with ρ^{2k} of its non-centrality, so $\hat{\rho}$ converts directly into a statement about power: at $\hat{\rho} = 0.7$ a quadratic departure retains about a quarter of its non-centrality and a cubic departure about an eighth. Any diagnostic that groups or bins on a fitted index can compute $\hat{\rho}$ from its own fit and read off the same limit. It equals 0.94 in design A, 0.71 at $\kappa = 0.10$, 0.69 at $\kappa = 0.25$ and 0.92 on the glaucoma data. Two approximations are involved in using it. It is the correlation of the adjusted index m_i with the truth, while the groups are formed on $\hat{\eta}_i$, which differs from m_i by the self-influence term of Section 4.3; and Proposition 2 concerns Hermite components of a continuous predictor, while the EDGE basis is a set of orthogonal polynomials over G discrete group means, so “zero otherwise” in the corollary holds up to the error of that discretisation. Both approximations improve as G grows and as the self-influence falls.

Corollary 1 *Among EDGE bases of degree $k \leq 3$ referred to the same null law, the retained non-centrality of a degree- j departure is ρ^{2j} if $j \leq k$ and zero otherwise, while the basis spends k degrees of freedom. Keeping the cubic direction only when $\hat{\rho}^6 \geq \tau$, for a threshold $\tau \in (0, 1)$, therefore retains it exactly when its signal survives at fraction τ . With $\tau = 0.2$ the rule takes $k = 3$ when $\hat{\rho} \geq 0.76$ and $k = 2$ otherwise, and we keep the cubic in addition whenever $\kappa < 0.05$, where there is no selection term to avoid.*

Proof By Proposition 2 the degree- j Hermite component of a departure survives grouping on the fitted predictor with amplitude ρ^j , so the non-centrality it contributes in the direction of the degree- j orthogonal polynomial is ρ^{2j} . An EDGE basis of degree k contains that direction when $j \leq k$ and is orthogonal to it otherwise, up to the discretisation error noted above, and it spends one degree of freedom on each of its k directions. Keeping the cubic direction when $\hat{\rho}^6 \geq \tau$ therefore keeps it exactly when the fraction ρ^6 of the cubic non-centrality that survives grouping is at least τ . $\square$

The value of the corollary is in the second consequence of Fitted relation 1: when the index is noisy the rule drops the cubic direction, which by Proposition 2 retains about $\rho^6 \approx 0.12$ of any cubic departure and by Fitted relation 1 carries almost all of the selection non-centrality. One decision removes a powerless direction and a biased one. Two guards are needed, and both exist because $\hat{\rho}$ is a null quantity: under a misspecified model the representation (9) reads the misfit as noise and $\hat{\rho}$ falls, so a rule that used $\hat{\rho}$ alone would discard directions precisely when a departure occupies them. The first guard is the floor at two. A degree-one basis is the calibration-slope direction alone and cannot see any change of shape, and in the simulations of Section 5.6 $\hat{\rho}$ fell from 0.66–0.71 under the null to 0.44–0.60 under the strongest departures, which without a floor would have cost the quadratic direction. The second is the aspect-ratio clause. At fixed p the selection term is negligible, so there is nothing to gain by dropping the cubic, and $\hat{\rho}$ falls from 0.93 under the null to 0.75 under a strong cubic departure — across the threshold, and therefore into discarding the direction that carries the signal, at a cost of 0.60 in power (Section 5.7). Keeping $k = 3$ whenever $\kappa < 0.05$ removes that failure without affecting any proportional-regime cell. What remains is a single decision, whether to keep the cubic when the index is noisy, and the rule makes it correctly on the glaucoma data of Section 6, where $\hat{\rho} = 0.92$ and the misfit lies in the cubic direction. Table 4 reports the sensitivity to τ : values below 0.2 keep the cubic too often and the size rises to 0.61–0.86 in the design where the selection term is largest, while $\tau = 0.3$ behaves like $\tau = 0.2$ and would still keep the cubic on the glaucoma data, where $\hat{\rho}^6 = 0.61$.

We call the result the *adaptive* EDGE basis and write EDGE- ρ where a symbol is needed.

Table 1. CALM in full. The inputs are one penalised fit and the analyst's own grouping; there is no resampling, and every step is a closed-form computation on quantities the fit already provides. The right-hand column names the result that licenses the step.

<i>Input:</i> design X with intercept column, responses $\mathbf{y}$, penalty λ , number of groups G , threshold $\tau = 0.2$.		
1. Fit the penalised model; form $\hat{\beta}$, $\hat{\pi}$, W , F , K and M , the G equal-frequency groups of $\hat{\pi}$ with C and V , and U and $\mathbf{r}$.	Section 2.1	
2. Debias in one step, $\tilde{\beta} = \hat{\beta} + F^{-1}K\hat{\beta}$; form $\mathbf{v} = \mathbf{r} - UM^{-1}K\tilde{\beta}$ and $A^* = V^{-1/2}C - UF^{-1}X^\top$.	(3)	
3. Centre the design; form $\psi = \mathbf{y} - \hat{\pi}$, $\hat{\gamma}$, $(\hat{\alpha}, \hat{\sigma})$ and $m_i = \mathbf{x}_i^\top \hat{\beta} - \hat{\gamma}\psi_i$.	(9)	
4. Estimate $(\hat{\alpha}, \hat{\sigma})$ by maximising the marginal likelihood under (10); form $\hat{w}_{0i}$ and $\Sigma_{\text{dn}} = A^* \hat{W}_0 A^{*\top}$.	Proposition 1	
5. Compute the index correlation $\hat{\rho}$ and take $k = 3$ if $\hat{\rho}^6 \geq \tau$ or $\kappa < 0.05$, and $k = 2$ otherwise; the decile basis instead spends all G directions.	Corollary 1	
6. Inflate the reference to $(1 + \hat{g})\Sigma_{\text{dn}}$, with $\hat{g} = 0.24\kappa$ on the decile basis, 1.15κ on EDGE-3 and 0.11κ on EDGE of degree at most two.	Fitted relation 1	
7. Refer SC.HL or SC.EDGE to the weighted chi-squared law with that covariance, evaluated by the algorithm of Davies (1980).	Section 2.3	
<i>Output:</i> the p -value for H_0^{str} along the fitted index, and the degree k that step 5 selected.		

4.5. Definition of CALM

CALM refers the shrinkage-corrected statistics of (4) to the exact weighted chi-squared law of Section 2.3 with covariance $(1 + \hat{g})\Sigma_{\text{dn}}$, where Σ_{dn} is the de-noised reference of Section 4.2, $\hat{g}$ is the basis-specific non-centrality fraction of Fitted relation 1 (0.24κ for the decile basis, 1.15κ for EDGE-3 and 0.11κ for EDGE of degree at most two), and the EDGE basis is the adaptive one of Corollary 1 together with the aspect-ratio clause that follows it. It requires one penalised fit and no resampling. At fixed p the inflation is negligible ($\hat{g} = 0.002$) and CALM is the reference (8). Table 1 states the whole procedure, and names the result that licenses each step.

An implementation of Table 1 as a single function, `calm.gof()`, is in the reproduction archive (Ebrahim, 2026b).

We write CALM-E for the test on the adaptive EDGE basis and CALM-D for the test on the decile basis, and use those names wherever the two differ.

The two bases are not on the same footing in the proportional regime, and Section 5.4 shows why this matters. The adaptive EDGE basis removes the selection term by dropping the direction that carries it, so the inflation is inert there while the rule takes degree two, as it does in every proportional-regime cell here: the factor is then at most 1.05, and removing it altogether changes the size by at most 0.008 in every cell we examined (Section 5). We keep it in the definition for uniformity across bases, and the statistic would behave the same without it. The decile basis spends all G directions, so it cannot avoid that direction and must instead absorb it into the reference through $\hat{g}$, a quantity that was fitted on eight cells. We therefore recommend the adaptive EDGE basis as the primary statistic and report the decile statistic, which is the shrinkage-corrected Hosmer–Lemeshow test and the one a reader of that tradition will look for, alongside it. The recommendation is not a preference between two equally good options. It is the one design in which holding the level does not depend on a constant fitted to reproduce the selection term, and Section 5.4 shows that this is what survives designs the constants never saw. It is not free of tuning: the degree rule carries the threshold τ , whose value is fixed in Section 4.4 and whose influence on the size is reported in Table 4.

5. Numerical results

5.1. Designs

We use the registered designs of Ebrahim (2026a) so that every number can be compared with that paper. Design A has $n = 500$, $p = 5$, independent standard normal covariates, $\beta = (0.5, -0.4, 0.3, -0.3, 0.2)^\top$ and $\lambda \in \{100, 137, 200\}$. Design B has $n = 400$, an autoregressive covariate correlation of 0.7, coefficients cycling $(0.35, -0.30, 0.25, -0.20, 0.15)$ scaled to $\|\beta\| = 2.31$, and $p = 100$ ($\kappa = 0.25$) with $\lambda \in \{50, 416, 1000\}$; a κ sweep keeps $\lambda = 416$ and takes $p = 40$ ($\kappa = 0.10$) and $p = 160$ ($\kappa = 0.40$). Under that scaling the standard deviation of the linear predictor is 1.48 to 1.51 across the three dimensions. A second scaling of the same coefficients is used

Table 2. Rejection rate of a correctly specified model at nominal level 0.05 under four references, eight cells: the first-order reference is conservative everywhere, the variance at the debiased fit is liberal at $\kappa \geq 0.25$, the de-noised variance repairs the lightest-penalty cell, and CALM holds the level on the decile basis and is conservative on the EDGE basis in the proportional regime. The first three EDGE columns use EDGE-3; the CALM columns use the adaptive EDGE basis and the basis-specific inflation of Section 4.5. 500 replications per cell.

Design	λ	Decile basis (SC.HL)				EDGE basis (SC.EDGE)			
		first order	Σ_d	Σ_{dn}	CALM-D	first order	Σ_d	Σ_{dn}	CALM-E
A ($\kappa = 0.01$)	100	0.030	0.044	0.046	0.046	0.028	0.050	0.048	0.050
A	137	0.022	0.036	0.040	0.038	0.030	0.040	0.038	0.040
A	200	0.018	0.044	0.046	0.046	0.030	0.050	0.048	0.048
B, $\kappa = 0.10$	416	0.006	0.054	0.066	0.060	0.012	0.048	0.054	0.040
B, $\kappa = 0.25$	50	0.020	0.086	0.046	0.032	0.030	0.120	0.056	0.006
B, $\kappa = 0.25$	416	0.018	0.076	0.072	0.052	0.024	0.080	0.076	0.028
B, $\kappa = 0.25$	1000	0.014	0.072	0.052	0.036	0.034	0.082	0.072	0.030
B, $\kappa = 0.40$	416	0.022	0.128	0.054	0.036	0.036	0.108	0.084	0.018

from Table 3 onwards and we write design B' for it: there the coefficients are scaled so that the standard deviation of the linear predictor, rather than $\|\beta\|$, equals 2.31, which makes $\|\beta\|$ equal to 3.54 to 3.61 and the signal about half again as strong. The two are not interchangeable, and we say which is used wherever it matters: Table 2, Tables 7, 8 and 9, Figure 4 and the supplementary material report design B, and the constants of Fitted relation 1 were fitted there; Tables 3, 4, 5 and 6, and the scale-recovery figures of Section 4.2, report design B'. All fits are ridge logistic regression by penalised iteratively reweighted least squares with the intercept unpenalised; $G = 10$ throughout. Under the null, data are generated from the design's own model, so every rejection is a false one. Sizes are based on 500 replications unless stated, for a Monte Carlo standard error of 0.010 at the 5 per cent level. Code that reproduces every number, table and figure below, with the random seeds, is in the reproduction archive (Ebrahim, 2026b).

5.2. Size

Table 2 gives the size of the four references. The first-order reference is conservative everywhere and severely so at $\kappa = 0.10$, where it rejects 0.6 per cent of correct models. The variance at the debiased fit holds its level at fixed p and at $\kappa = 0.10$ and is liberal at $\kappa \geq 0.25$. The de-noised variance repairs the lightest-penalty cell and halves the excess at $\kappa = 0.40$. CALM holds the level on the decile basis, where every cell is within Monte Carlo error of the nominal level, and never exceeds it on the EDGE basis: there the rule of Corollary 1 drops the cubic direction in the proportional regime, and the two-dimensional basis rejects 0.6 to 4 per cent of correct models, least at $\kappa = 0.25$, $\lambda = 50$. That conservatism is not caused by the inflation, and it is not an artefact of the level: Section S3 of the supplementary material repeats the check at 0.01 and 0.10, where the adaptive statistic rejects between 0.000 and 0.062 and between 0.030 and 0.154 of correct models. On the EDGE basis the factor is at most 1.05 while the rule takes degree two, as it does in every proportional-regime cell here, and removing it altogether changes the size by at most 0.008 in every cell we examined; the de-noised reference overstates the null variance in the low-degree directions while understating it in the cubic direction, and dropping the cubic leaves only the overstated part. We keep the inflation on this basis for consistency of definition rather than for its effect, and the statistic would behave the same without it.

5.3. Robustness to the design law

Assumption 2 requires Gaussian predictors. Table S2 of the supplementary material repeats three cells with predictors drawn from a t_3 distribution scaled to unit variance and from a standardised Bernoulli(0.3) law, and finds the size of the de-noised reference between 0.042 and 0.098, the worst being the heavy-tailed design, where the level doubles. Gaussianity is a condition of the proof rather than a hard boundary of the method, but the heavy-tailed cells are consistently the least reliable. Section 5.4 goes further and runs the whole procedure under those two laws.

5.4. Validation outside the calibration cells

The constants of Fitted relation 1 were fitted on the eight cells of Table 2, and Table 2 therefore reports the size of CALM in the cells that calibrated it. Table 3 reports it in seventeen further cells: two intermediate aspect ratios, two penalties at $\kappa = 0.40$, a design with covariate correlation 0.3 instead of 0.7, a weaker and a stronger signal than design B', a design twice the size at the same aspect ratio, twenty groups instead of ten, and the three predictor laws of Section 5.3 run through the whole procedure rather than through Σ_{dn} alone. Sixteen of the seventeen did not calibrate the constants; the exception is the $\text{sd}(\eta) = 1.5$ row, which is design B at $\lambda = 416$ up to the scaling, and which we report at an independent seed as a check on Monte Carlo error rather than as a held-out cell.

The two bases behave differently, and the difference is the main practical finding of this section. On the adaptive EDGE basis the size lies between 0.024 and 0.080 in every one of the seventeen cells, with fifteen of them within three Monte Carlo standard errors of the nominal level and the largest at 0.080; the level there does not rest on a constant fitted to the selection term, because the rule of Corollary 1 drops the direction that term occupies rather than correcting for it. On the decile basis, which spends all G directions and must therefore absorb that term into the reference through $\hat{g}$, the size is between 0.034 and 0.076 in twelve cells and fails in the remaining five, most severely at covariate correlation 0.3, where it rejects 0.688 of correct models.

Where the decile correction fails, and why. The failures have one cause, and it is visible in Fitted relation 1 itself: the selection term is not a function of κ . An observation changes group when its own response moves its fitted index across a group boundary, so the term grows with the self-influence $\bar{h}$ measured against the spread of the fitted index, and with the number of observations per group. At $\kappa = 0.25$ the squared mean $\|\mathbb{E}v\|^2$ is nine times larger under covariate correlation 0.3 than under 0.7, because the fitted index has half the spread there while the self-influence is unchanged, and it doubles when n doubles at fixed κ . The quantity

$$\hat{q} = n\bar{h}^2/\text{sd}(\hat{\eta})^2, \quad (13)$$

which the boundary-crossing argument suggests and which we call the crossing index, tracks the term across these designs better than κ does but not closely enough to replace it, and a per-dataset estimate built from the leave-one-out displacement of the index proved far too noisy to use. The open problem is therefore precise: is there an observable, built from the self-influence and the spread of the fitted index, that predicts $\|\mathbb{E}v\|^2$ well enough to replace the constant 0.24κ across design families? Until there is, the decile reference should be read as calibrated for designs resembling those of Table 2. The prepivoting bootstrap is not a way out: in the two failing designs we also ran it, its size is 0.11 and 0.07, so the difficulty is in the corrected statistic and not in the analytic reference alone. The obstruction is therefore a property of grouping on a noisy index, not of the reference proposed here, and the comparison of the two bases is itself the finding: a statistic that spends every direction has to model the selection term, while a statistic that chooses its directions can avoid it. Where a diagnostic has that freedom, avoiding the contaminated direction proved more robust than correcting for it wherever that direction stays put, which under balanced outcomes it does, because the correction needs a constant fitted on particular designs and the avoidance needs none; Section 5.5 reports the regime in which the direction moves and the ordering of the two bases reverses. We record the transfer of that correction as the outstanding problem for anyone who wants the test in its Hosmer–Lemeshow form rather than its smooth one.

5.5. Unbalanced outcomes, smaller samples and other groupings

Three design families lie wholly or partly outside Table 3 and a reader of the clinical prediction literature will ask about all three. Table 6 reports them, and the first is a boundary of the method rather than a further confirmation of it.

Outcomes here have been balanced. Shifting the intercept breaks both statistics in the proportional regime: at $\kappa = 0.25$ the decile statistic rejects 0.060, 0.204 and 0.492 of correct models at prevalences 0.30, 0.15 and 0.08, and the smooth statistic 0.140, 0.574 and 0.884: the smooth statistic loses its level first, already at a prevalence of 0.30. At fixed dimension the decile statistic is unaffected, at 0.054 and 0.050, but the smooth one is not: its size rises to 0.088 at prevalence 0.15 and to 0.130 at prevalence 0.08, eight Monte Carlo standard errors above the nominal level, so the loss there is much smaller than in the proportional regime without being absent. The order is the reverse of the balanced case, where the smooth basis was the reliable one. The cause is the selection term of Section 4.3, and it can be isolated. Let the ratio $\mathbb{E}(S)/\mathbb{E}\{\text{tr}(P_Z \Sigma_{\text{dn}})\}$, on the adaptive EDGE basis and with Σ_{dn} un-inflated,

Table 3. Rejection rate of a correctly specified model at nominal level 0.05 in designs that were not used to calibrate the constants of Fitted relation 1: nine designs with the Gaussian predictor law and eight repetitions of earlier cells under t_3 and standardised Bernoulli predictors. The last column is the mean of the observable $\hat{\rho}$. 500 replications per cell; the Monte Carlo standard error is 0.010. Design B' of Section 5.1 throughout.

Design	CALM-D	CALM-E	EDGE-3	$\hat{\rho}$
<i>New designs, Gaussian predictors</i>				
$\kappa = 0.15$ ($p = 60$)	0.066	0.054	0.080	0.73
$\kappa = 0.33$ ($p = 132$)	0.052	0.030	0.060	0.72
$\kappa = 0.40$, $\lambda = 50$	0.130	0.024	0.138	0.70
$\kappa = 0.40$, $\lambda = 1000$	0.034	0.056	0.062	0.73
$\kappa = 0.25$, correlation 0.3	0.688	0.076	0.872	0.70
$\kappa = 0.25$, $\text{sd}(\eta) = 1.5$	0.054	0.028	0.030	0.69
$\kappa = 0.25$, $\text{sd}(\eta) = 3.0$	0.082	0.080	0.132	0.74
$\kappa = 0.25$, $n = 800$, $p = 200$	0.122	0.074	0.168	0.74
$\kappa = 0.25$, $G = 20$	0.044	0.056	0.102	0.73
<i>Other predictor laws</i>				
A, t_3	0.050	0.050	0.050	0.92
A, binary	0.052	0.058	0.058	0.94
$\kappa = 0.10$, t_3	0.112	0.080	0.164	0.72
$\kappa = 0.10$, binary	0.064	0.074	0.094	0.73
$\kappa = 0.25$, t_3	0.066	0.044	0.084	0.72
$\kappa = 0.25$, binary	0.054	0.044	0.072	0.72
$\kappa = 0.40$, t_3	0.076	0.050	0.094	0.71
$\kappa = 0.40$, binary	0.056	0.044	0.078	0.72

Table 4. Sensitivity of the size of the adaptive EDGE statistic to the threshold τ of Corollary 1, at nominal level 0.05, in seven designs. The value used throughout is $\tau = 0.2$. 500 replications per cell; the $\tau = 0.2$ column is an independent run from the one behind the CALM-E column of Table 2, and the two differ by one to two Monte Carlo standard errors. Design B' of Section 5.1 throughout.

Design	threshold τ			
	0.05	0.10	0.20	0.30
A ($\kappa = 0.01$)	0.050	0.050	0.050	0.050
B, $\kappa = 0.10$	0.070	0.070	0.052	0.040
B, $\kappa = 0.25$	0.062	0.064	0.044	0.032
B, $\kappa = 0.40$	0.048	0.040	0.032	0.024
$\kappa = 0.40$, $\lambda = 50$	0.138	0.092	0.024	0.022
$\kappa = 0.25$, correlation 0.3	0.856	0.610	0.076	0.024
$\kappa = 0.10$, t_3	0.162	0.156	0.080	0.048

be the quantity the reference has to reproduce, equal to one when the reference is correct, and split it into the part contributed by the selection mean $\mathbf{m} = \mathbb{E}(\mathbf{v})$ and a remainder. Table 5 reports the split. The remainder stays between 0.88 and 1.00 as the prevalence falls from 0.50 to 0.08, so the de-noised variance of Section 4.2 remains accurate throughout, including where $\hat{\rho}$ has fallen to 0.62; the entire excess is non-centrality, whose share grows from 0.24 to 8.91 against the inflation of 0.03 that the κ -law supplies.

Two properties of the selection mean move together, and between them they say why the smooth basis fares worse. Its squared length grows from 0.59 to 5.05. Its shape also turns. Under balanced outcomes the displacement is the odd, cubic curve that the self-influence argument predicts, and the quadratic basis retains two per cent of it, which is precisely why the degree rule of Section 4.4 discards the cubic in the proportional regime and why the inflation constant at degree two is a tenth of the one at degree three; at prevalence 0.08 the displacement is a U-shape and the quadratic basis retains ninety-nine per cent. The rule is calibrated on the premise that the term to be avoided is cubic, and that premise is a property of balanced outcomes rather than of the method.

Three repairs were measured against these numbers. Building the polynomials on the group-mean linear predictor rather than the group-mean probability, so that lowering the prevalence becomes a translation of the index and the

Table 5. Where the excess comes from, at $\kappa = 0.25$. The ratio $\mathbb{E}(S)/\mathbb{E}\{\text{tr}(P_Z \Sigma_{\text{dn}})\}$ that the reference has to reproduce is one when the reference is correct; it is split here into the part contributed by the selection mean $\mathbf{m} = \mathbb{E}(\mathbf{v})$ and the remainder, so a remainder near one locates the whole error in the non-centrality. The last column is the fraction of $\|\mathbf{m}\|^2$ that a degree-two basis retains. 500 replications per row. Design B' of Section 5.1 throughout.

Prevalence	CALM-D	CALM-E	Remainder	Selection share	$\ \mathbf{m}\ ^2$	In basis
0.50	0.056	0.044	0.88	0.24	0.59	0.02
0.30	0.060	0.140	0.95	0.98	0.95	0.43
0.15	0.204	0.574	1.00	4.05	2.40	0.86
0.08	0.492	0.884	0.93	8.91	5.05	0.99

Table 6. Rejection rate of a correctly specified model at nominal level 0.05 in three design families that Table 3 does not cover, or covers only in part: unbalanced outcomes, smaller samples at the same aspect ratio, and other numbers of groups. The $\kappa = 0.25$, $G = 20$ cell is repeated from Table 3 so that the G family can be read in one place. 500 replications per cell. Design B' of Section 5.1 throughout.

Design	CALM-D	CALM-E	$\hat{\rho}$
$\kappa = 0.25$, prevalence 0.30	0.060	0.140	0.72
$\kappa = 0.25$, prevalence 0.15	0.204	0.574	0.68
$\kappa = 0.25$, prevalence 0.08	0.492	0.884	0.62
A, prevalence 0.15	0.054	0.088	0.85
A, prevalence 0.08	0.050	0.130	0.69
$\kappa = 0.25$, $n = 200$	0.054	0.018	0.70
$\kappa = 0.25$, $n = 150$	0.038	0.038	0.68
A, $n = 200$	0.056	0.028	0.84
$\kappa = 0.25$, $G = 5$	0.068	0.054	0.73
$\kappa = 0.25$, $G = 20$	0.044	0.056	0.73
A, $G = 5$	0.054	0.058	0.93
A, $G = 20$	0.058	0.042	0.93

compression of the fitted probabilities disappears, raises the size in every unbalanced cell, to 0.928 against 0.884 at prevalence 0.08: the displacement is smooth on either scale and the basis retains it either way. Reducing the number of groups does not help either, the size at prevalence 0.08 being 0.790, 0.884 and 0.916 at $G = 5$, 10 and 20. What does work is to form the groups from a function of the predictors alone, which carries no selection by construction: the size returns to 0.034 and 0.066 and the ratio to 1.00 and 1.18. That is a diagnosis rather than a proposal, and the same run says why. Grouping on the first principal component of X instead of the fitted index costs more than half the power against a quadratic departure in the designs where both groupings are valid, 0.215 against 0.488 at prevalence 0.50.

We report the boundary rather than exclude the designs, and it is worth being exact about what lies inside it. What fails is the response-dependent grouping inherited from the Hosmer–Lemeshow construction, not the reference, which the third repair shows is still accurate at prevalence 0.08. The method as it stands is for outcomes that are not strongly unbalanced once p/n is an appreciable fraction; few events in the proportional regime need a grouping whose selection term is controlled, and that is the undetermined constant of Section 4.3 met again in the regime where it stops being small. At fixed dimension the same term is far weaker but not zero: the decile statistic holds its level at prevalence 0.08, at 0.050, while the smooth one rejects 0.130, so a strongly unbalanced outcome warrants the same caution there in milder form.

The other two families raise no difficulty. Halving the sample at the same aspect ratio, to $n = 200$ and $n = 150$ with $p = 50$ and $p = 37$, leaves the size between 0.018 and 0.056, so the proportional-regime machinery does not need a large sample so much as a stable ratio. Changing the number of groups to $G = 5$ and $G = 20$ leaves it between 0.042 and 0.068; the selection term grows with the number of observations per group and shrinks with their number, and over this range the two effects offset.

5.6. Power

An exact reference has no Monte Carlo error in its p -value, and a bootstrap with B refits does; one may therefore hope for a small gain in power at no cost in size. But the more useful question is how CALM compares with what an analyst would otherwise use, and Figure 3 and Table 7 answer it on identical simulated datasets in the four κ cells, against two departures a grouped test should be able to see: a Hermite quadratic along the true index and an omitted interaction x_1x_2 off the index, each at 0.5, 1 and 1.5 times the standard deviation of the linear predictor for the first and 1 and 1.5 times for the second. The comparison set is the prepivoting bootstrap of Ebrahim (2026a) with 399 refits, the generalised residual prediction (GRP) test of Janková et al. (2020) with its own lasso fit (R package `GRPtests` 0.1.2, archived on the Comprehensive R Archive Network since May 2022 and installed from the archived tarball; default settings: five sample splits, lasso tuned by cross-validation), the classical Hosmer–Lemeshow test on the unpenalised maximum likelihood fit where that fit exists, and the four checks a clinical prediction paper would report on the ridge fit itself: the Hosmer–Lemeshow test without correction, the calibration intercept-and-slope test (Cox, 1958), Spiegelhalter's z (Spiegelhalter, 1986) and the unweighted sum of squares of Copas (1989), together with the Hosmer–Lemeshow test on ten-fold cross-validated predictions. A test's power curve is drawn only in cells where it holds its level; a rejection rate above a broken level is not power, and those tests appear in Table 8 with their size and their size-adjusted power, the rejection rate at the critical value that would give the test its nominal size in that cell. Every check a clinical prediction paper would report on the ridge fit rejects almost every correctly specified model, at fixed p as well as in the proportional regime, because each of them is displaced by shrinkage in the way Section 2 describes for the Hosmer–Lemeshow statistic; and even size-adjusted, the calibration-slope and Spiegelhalter statistics do not rank the departures above the null, because a quadratic departure moves the apparent slope back towards one; the uncorrected Hosmer–Lemeshow statistic keeps its ranking power at fixed p (0.84) and loses it in the proportional regime. The test on cross-validated predictions rejects for a different reason: shrunk predictions are miscalibrated on new data even when the model is correct. Under the correct model the calibration slope of the ten-fold predictions is 1.77 in design A and 2.25 and 2.00 at $\kappa = 0.10$ and 0.25, against the value of one that calibration would require, so the predictions are too flat out of sample as well as in it, and the test detects that rather than any error in the model.

Three things are visible. First, on the EDGE basis CALM loses no power against the bootstrap. The paired differences against the quadratic departure are within ± 0.05 at every κ (0.640 against 0.632 at fixed p and relative size 0.5; 0.708 against 0.695 and 0.895 against 0.875 at $\kappa = 0.10$; 0.420 against 0.458 and 0.640 against 0.615 at $\kappa = 0.25$; 0.077 against 0.125, 0.260 against 0.295 and 0.350 against 0.367 at $\kappa = 0.40$), with 399 refits replaced by one fit; the largest, 0.048 at $\kappa = 0.40$ and relative size 0.5, is the price of the conservative size there, 0.018 against 0.050, and taken at each test's own empirical five per cent point the rates are 0.153 against 0.125. On the decile basis CALM matches the bootstrap at fixed p and at $\kappa = 0.10$ and gives up 0.10–0.12 at $\kappa = 0.25$ (0.233 against 0.333, 0.310 against 0.432); about half of that gap is the inflation $(1 + 0.24\kappa)$ that protects its level and half is the de-noised variance itself, since the un-inflated reference Σ_{dn} gives 0.280 and 0.368 on the same data; at $\kappa = 0.40$ the bootstrap's higher rates (0.237, 0.302) sit on a size of 0.074. Second, CALM recovers the classical power. At fixed p the Hosmer–Lemeshow test on the maximum likelihood fit rejects the quadratic departure at 0.390, 0.983 and 0.997; CALM on the ridge fit, after subtracting the displacement and correcting the reference, rejects it at 0.343, 0.965 and 0.995, and its EDGE basis at 0.640, 0.998 and 1. Third, CALM and the test of Janková et al. (2020) see different things, as Ebrahim (2026a) found in a paired simulation against the same `GRPtests` implementation, where the two rejected on essentially disjoint sets of datasets. The generalised residual prediction test is the stronger test off the index — 0.457 and 0.897 against the interaction at fixed p , where CALM's EDGE basis gives 0.352 and 0.517 and no grouped test has power once $\kappa > 0$ — and the weaker along it, with 0.020, 0.407 and 0.877 against the quadratic at fixed p and 0.010, 0.083 and 0.297 at $\kappa = 0.10$, where CALM gives 0.233, 0.708 and 0.895. It also keeps some of its off-index power in the proportional regime (0.247–0.420 at $\kappa = 0.25$, 0.253–0.340 at $\kappa = 0.40$), where the interaction is absorbed by the ridge fit and no grouped test sees it. One test asks whether the model is right as a function of the covariates, the other whether the deployed index is calibrated along itself; an analyst who wants both should run both.

Designs A and B are dense, while the test of Janková et al. (2020) assumes a sparse coefficient vector, so the comparison above is made outside its own assumptions and its conservatism there (0.003 to 0.043 under the null) may be a consequence of that. We therefore repeated the comparison at $\kappa = 0.25$ with only five non-zero coefficients out of 100, where it is on its own ground. Its size is then 0 in 300 replications, still conservative; against the off-index

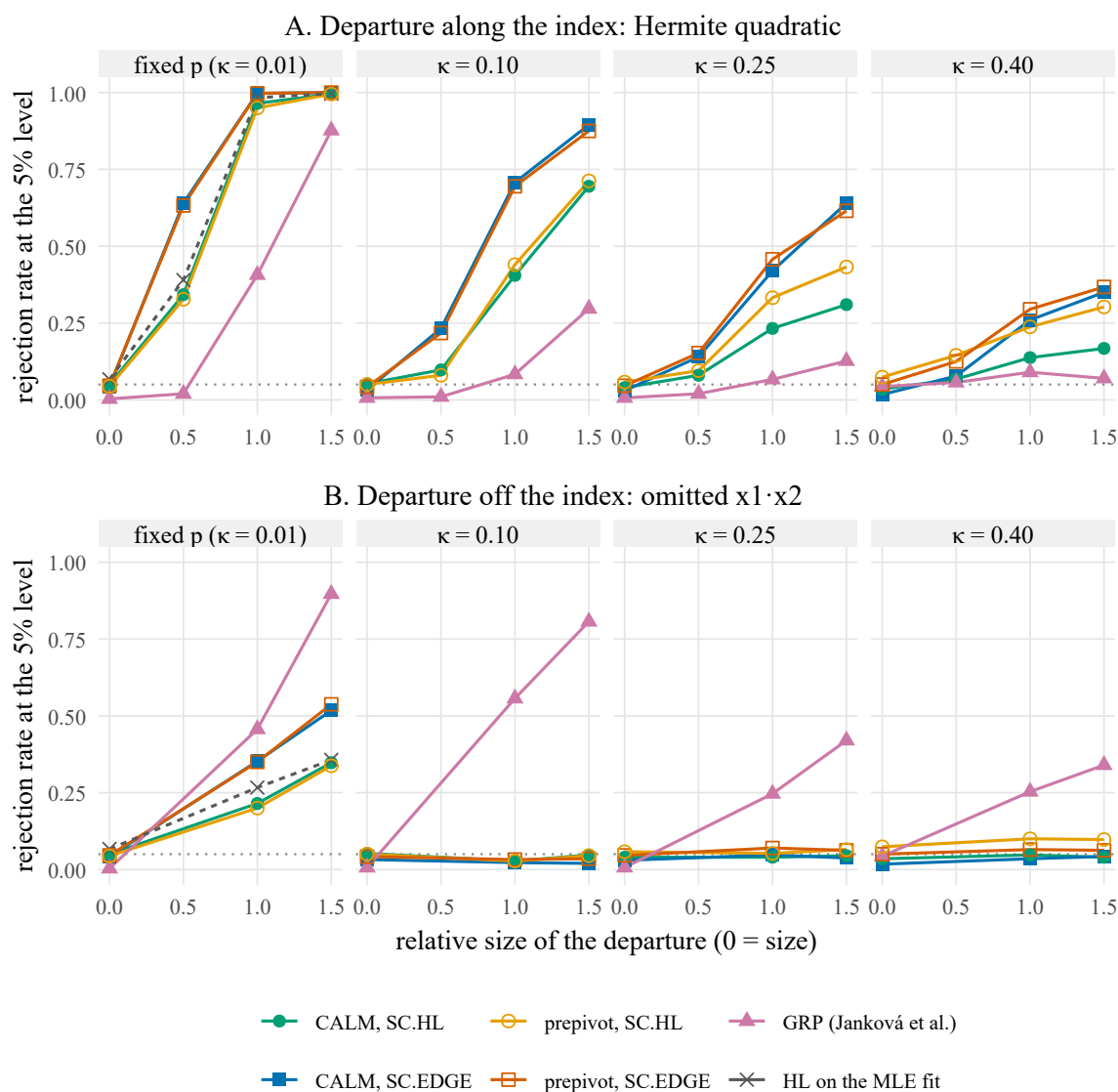


Figure 3 Power at nominal level 0.05 against a quadratic departure along the index (A) and an omitted interaction off the index (B), by relative size of the departure, in the four κ cells, on identical datasets: CALM on both bases, pre pivoting with 399 refits, the generalised residual prediction test and the Hosmer–Lemeshow test on the maximum likelihood fit. A curve is drawn only where the test holds its level in that cell; the tests that hold it nowhere are in Table 8. Relative size 0 is size. 300–500 replications per point.

interaction it rejects 0.87, against 0.03 for CALM, and against the quadratic along the index it rejects 0.077 against 0.21 for the adaptive EDGE basis. Sparsity therefore sharpens the division of labour rather than changing it: the covariate-function test finds what is off the index, the calibration test finds what is along it. The classical test on the maximum likelihood fit, finally, is not an option in the proportional regime: it rejects 9.7 per cent of correct models at $\kappa = 0.10$ and 33 per cent at $\kappa = 0.25$, and at $\kappa = 0.40$ the maximum likelihood estimate did not exist in any of the 300 datasets (Candès and Sur, 2020).

Table 7. Power at nominal level 0.05 of the tests that hold their level, on identical datasets, against a quadratic departure along the index and an omitted interaction off the index. Rows with relative size 0 are size. Entries are omitted where the maximum likelihood estimate did not exist in most datasets or where the test does not hold its level in that cell. 300–500 replications per cell.

Design	departure	size	CALM		prepivot		MLE fit		
			CALM-D	CALM-E	SC.HL	SC.EDGE	GRP	HL	EDGE
A ($\kappa = 0.01$)	—	0	0.050	0.043	0.044	0.046	0.003	0.067	0.053
	quadratic in the index	0.5	0.343	0.640	0.328	0.632	0.020	0.390	0.650
	quadratic in the index	1	0.965	0.998	0.950	0.998	0.407	0.983	0.997
	quadratic in the index	1.5	0.995	1.000	0.995	1.000	0.877	0.997	1.000
	$x_1 x_2$ off the index	1	0.215	0.352	0.200	0.350	0.457	0.267	0.413
	$x_1 x_2$ off the index	1.5	0.347	0.517	0.338	0.537	0.897	0.357	0.560
B, $\kappa = 0.10$	—	0	0.052	0.033	0.050	0.042	0.007	0.097	0.067
	quadratic in the index	0.5	0.098	0.233	0.080	0.217	0.010	—	—
	quadratic in the index	1	0.405	0.708	0.440	0.695	0.083	—	—
	quadratic in the index	1.5	0.695	0.895	0.713	0.875	0.297	—	—
	$x_1 x_2$ off the index	1	0.028	0.022	0.028	0.033	0.557	—	—
	$x_1 x_2$ off the index	1.5	0.048	0.020	0.045	0.035	0.807	—	—
B, $\kappa = 0.25$	—	0	0.040	0.030	0.058	0.046	0.007	0.326	0.237
	quadratic in the index	0.5	0.080	0.140	0.095	0.152	0.020	—	—
	quadratic in the index	1	0.233	0.420	0.333	0.458	0.067	—	—
	quadratic in the index	1.5	0.310	0.640	0.432	0.615	0.127	—	—
	$x_1 x_2$ off the index	1	0.040	0.048	0.052	0.070	0.247	—	—
	$x_1 x_2$ off the index	1.5	0.045	0.037	0.065	0.062	0.420	—	—
B, $\kappa = 0.40$	—	0	0.035	0.018	0.074	0.050	0.043	—	—
	quadratic in the index	0.5	0.068	0.077	0.145	0.125	0.057	—	—
	quadratic in the index	1	0.138	0.260	0.237	0.295	0.090	—	—
	quadratic in the index	1.5	0.168	0.350	0.302	0.367	0.070	—	—
	$x_1 x_2$ off the index	1	0.048	0.035	0.100	0.065	0.253	—	—
	$x_1 x_2$ off the index	1.5	0.040	0.043	0.098	0.062	0.340	—	—

Table 8. Calibration checks applied to the ridge fit: rejection rate of a correctly specified model (size) and size-adjusted power against the quadratic departure of relative size 1, where the critical value is the empirical 5 per cent point of the same test's null distribution in the same cell. A dash marks cells in which every null p-value was below 10^{-10} , so that no empirical critical value exists.

Test	A		$\kappa = 0.10$		$\kappa = 0.25$		$\kappa = 0.40$	
	size	adj. power	size	adj. power	size	adj. power	size	adj. power
uncorrected Hosmer–Lemeshow	0.933	0.840	1.000	0.093	1.000	—	1.000	—
calibration intercept and slope	1.000	0.000	1.000	0.000	1.000	—	1.000	—
Spiegelhalter's z	1.000	0.003	1.000	0.000	1.000	0.000	1.000	—
unweighted sum of squares	1.000	—	1.000	—	1.000	—	1.000	—
Hosmer–Lemeshow on cross-validated predictions	0.593	0.873	0.717	0.140	0.420	0.080	0.250	0.077
Hosmer–Lemeshow on the MLE fit	0.067	0.970	0.097	0.820	0.326	0.197	—	—

5.7. The basis degree and the attenuation law

Figure 4 tests Corollary 1 directly. Against a degree-two Hermite departure along the index, the degree-matched basis EDGE-2 has more power than EDGE-3 in every cell and the difference grows with κ — 0.510 against 0.492 at $\kappa = 0.25$ and rel = 1, 0.704 against 0.650 at rel = 1.5 — while both leave the decile basis far behind (0.360). Against a degree-three departure, only EDGE-3 among the EDGE bases has power at fixed p (0.558, against 0.050 for EDGE-2 and 0.356 for the decile basis), and at $\kappa \geq 0.10$ no basis has any (0.04–0.09), as $\hat{\rho}^6$ predicts: it is 0.12 under the null in those cells and falls to 0.06 under the departure itself. The observable $\hat{\rho}$ is 0.94, 0.71 and 0.69 in

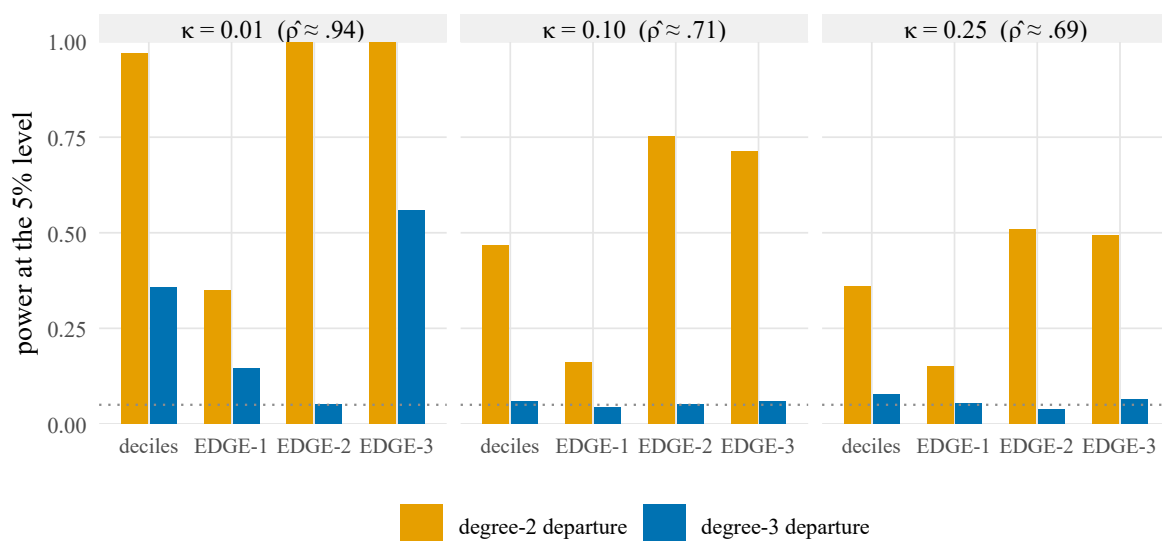


Figure 4 Power at nominal level 0.05 of the decile basis and of EDGE bases of degree one to three against degree-two and degree-three Hermite departures along the index of size $\text{rel} = 1$, at $\kappa = 0.01, 0.10$ and 0.25 . 500 replications per cell.

the three cells under the null; the rule of Corollary 1 selects degree three, two and two, which is the most powerful choice in every row of the figure.

The aspect-ratio clause of the rule can be seen at work in a separate run of 500 replications in the same three cells, on design B' , so that its rejection rates are not comparable cell for cell with those of Table 7, which are on design B. Against a cubic departure at fixed p , $\hat{\rho}$ falls from 0.93 under the null to 0.84 at relative size 1 and 0.70 at relative size 1.5. A rule that used $\hat{\rho}$ alone would then keep the cubic in only 88 and 44 per cent of replications and reject at 0.496 and 0.372; keeping $k = 3$ whenever $\kappa < 0.05$ raises those to 0.542 and 0.666, matching the fixed EDGE-3 basis exactly. In the two proportional-regime cells the two rules agree in every replication, so the clause costs nothing there. Against a quadratic departure the adaptive basis rejects at 0.916 and 0.980 at $\kappa = 0.10$ and at 0.690 and 0.846 at $\kappa = 0.25$, against 0.642 and 0.912, and 0.426 and 0.542, for the decile basis, which is the attenuation law of Proposition 2 read as a statement about power.

5.8. The boundary: departures that do not survive attenuation

Table 9 shows what the attenuation law forbids, and it is the reason the departures of Section 5.6 were chosen as they were. Against a cubic departure along the index and a small interaction off it, CALM and the bootstrap both hold their level at every κ , and neither has power once $\kappa > 0$: every rate in the proportional regime is within Monte Carlo error of the corresponding size, and the bootstrap's 0.098 and 0.104 at $\kappa = 0.40$ sit on a size of 0.074. This is Proposition 2 and not a property of either reference. With $\hat{\rho} \approx 0.7$ a cubic departure retains about $\rho^6 \approx 0.12$ of its non-centrality, so no grouped test on the fitted index can see it, and the interaction is absorbed by the ridge fit. At fixed p , where $\hat{\rho} \approx 0.9$, both tests see the cubic (0.304 and 0.322 on the EDGE basis at relative size 0.7) and CALM is again level with the bootstrap. What CALM offers in the proportional regime is therefore validity, with power against the departures that survive the index.

5.9. Other smooth penalties

Nothing in Sections 2–4 uses the linearity of the ridge penalty. If a smooth penalty contributes a score term $\mathbf{a}(\beta)$ with $K(\beta) = -\partial \mathbf{a} / \partial \beta$, then $\hat{\beta} = \hat{\beta} - F^{-1} \mathbf{a}(\hat{\beta})$, $\hat{\mu} = -UM^{-1} \mathbf{a}(\hat{\beta})$ and Theorem 1 go through with K evaluated at $\hat{\beta}$ (Ebrahim, 2026a, Remark 2). Section S1 of the supplementary material reports the generalised ridge penalty and Firth's penalty (Firth, 1993), the standard response to separation in clinical prediction (Puhr et al., 2017), whose

Table 9. Size and rejection rates at nominal level 0.05 of CALM and of pre pivoting with 399 refits, paired on identical datasets, against departures that do not survive attenuation — a Hermite cubic along the true index and a small omitted interaction x_1x_2 , each at relative size 0.4 and 0.7 of the linear predictor. Rows with relative size 0 are size. k is the EDGE degree chosen by Corollary 1 in the majority of replications; diff is the paired difference CALM minus pre pivot with its standard error. 500 replications per cell.

Design	departure	size	k	Decile basis (SC.HL)			Adaptive EDGE basis		
				CALM-D	pre pivot	diff (s.e.)	CALM-E	pre pivot	diff (s.e.)
A ($\kappa = 0.01$)	—	0	3	0.046	0.044	+0.002 (0.005)	0.050	0.046	+0.004 (0.005)
	cubic in the index	0.4	3	0.108	0.106	+0.002 (0.004)	0.116	0.118	−0.002 (0.006)
	cubic in the index	0.7	3	0.194	0.184	+0.010 (0.007)	0.304	0.322	−0.018 (0.008)
	x_1x_2 off the index	0.4	3	0.082	0.078	+0.004 (0.004)	0.112	0.108	+0.004 (0.007)
	x_1x_2 off the index	0.7	3	0.144	0.124	+0.020 (0.007)	0.250	0.258	−0.008 (0.009)
B, $\kappa = 0.10$	—	0	2	0.060	0.050	+0.010 (0.005)	0.040	0.042	−0.002 (0.009)
	cubic in the index	0.4	2	0.036	0.040	−0.004 (0.005)	0.020	0.022	−0.002 (0.007)
	cubic in the index	0.7	2	0.026	0.030	−0.004 (0.003)	0.020	0.040	−0.020 (0.008)
	x_1x_2 off the index	0.4	2	0.046	0.034	+0.012 (0.006)	0.032	0.044	−0.012 (0.007)
	x_1x_2 off the index	0.7	2	0.050	0.048	+0.002 (0.003)	0.044	0.052	−0.008 (0.011)
B, $\kappa = 0.25$	—	0	2	0.052	0.058	−0.006 (0.005)	0.028	0.046	−0.018 (0.010)
	cubic in the index	0.4	2	0.038	0.062	−0.024 (0.007)	0.026	0.036	−0.010 (0.007)
	cubic in the index	0.7	2	0.036	0.078	−0.042 (0.009)	0.026	0.032	−0.006 (0.008)
	x_1x_2 off the index	0.4	2	0.034	0.044	−0.010 (0.004)	0.028	0.044	−0.016 (0.010)
	x_1x_2 off the index	0.7	2	0.042	0.062	−0.020 (0.006)	0.022	0.058	−0.036 (0.011)
B, $\kappa = 0.40$	—	0	2	0.036	0.074	−0.038 (0.009)	0.018	0.050	−0.032 (0.009)
	cubic in the index	0.4	2	0.042	0.098	−0.056 (0.010)	0.012	0.050	−0.038 (0.009)
	cubic in the index	0.7	2	0.026	0.104	−0.078 (0.012)	0.020	0.048	−0.028 (0.010)
	x_1x_2 off the index	0.4	2	0.046	0.076	−0.030 (0.008)	0.024	0.062	−0.038 (0.010)
	x_1x_2 off the index	0.7	2	0.058	0.086	−0.028 (0.008)	0.032	0.062	−0.030 (0.011)

implementation we checked against the `logistf` package (Heinze and Schemper, 2002). The first behaves exactly as the ridge penalty does. The second marks a boundary of the framework: at fixed p Firth's correction is $O(n^{-1})$ and there is nothing to correct, but in the proportional regime it is not a shrinkage penalty at all, the fitted coefficient norm exceeding the true one, so the displacement that this paper removes has the wrong sign and the corrected references reject 12 to 32 per cent of correct models. The framework is for estimators that shrink towards zero.

6. The glaucoma model revisited

Ebrahim (2026a) analysed a ridge logistic model for glaucoma diagnosis from $p = 62$ morphometric measurements of the optic nerve head on $n = 196$ eyes (the GlaucomaM data; see Coan et al., 2023 for the clinical background), a design in which the maximum likelihood estimate does not exist (Candès and Sur, 2020) and $\kappa = 0.32$. At the cross-validated penalty the uncorrected Hosmer–Lemeshow test returned $p < 10^{-5}$; the pre pivoted corrected test returned 0.026 on the decile basis and 0.034 on the EDGE basis, and the paper concluded that the evidence for misfit was real but modest and that the model needed recalibration rather than rebuilding. Table 10 adds the analytic references. The first-order reference is conservative, as Theorem 1 predicts at this κ ; the variance at the debiased fit gives 0.024 and 0.034, within 0.002 of the bootstrap; CALM gives 0.012 and 0.033, with $\hat{\rho} = 0.92$ so that the adaptive rule keeps all three EDGE directions. The de-noised reference is the most liberal of the four, at 0.007 and 0.011, and the inflations of Section 4.5 bring the p-values back to 0.012 and 0.033: $1 + 0.24\kappa = 1.08$ on the decile basis and, because $\hat{\rho}$ is high enough here for the rule to keep the cubic, $1 + 1.15\kappa = 1.36$ on the EDGE basis. De-noising lowers the p-value here because the signal is strong and the index is accurate: the observables are $\hat{a} = 1.00$, $\hat{\sigma} = 0.43$ and $\hat{c} = 3.14$, giving an estimated standard deviation of 3.15 for the true linear predictor against 1.11 for the fitted one, so the estimated null Bernoulli variance is well below the fitted variance and $\text{tr}(\Sigma_{\text{dn}})$ is 0.82 of $\text{tr}(\Sigma_{\text{d}})$. Two cautions belong with the number. This dataset lies outside every simulated design: $\hat{\rho} = 0.92$ against 0.65 to 0.74 in the proportional-regime cells on which the constants were fitted, and the maximum likelihood

Table 10. The glaucoma model at the cross-validated penalty $\lambda = 295$ ($\lambda_g = 1.51$), $G = 10$: p -values of the uncorrected test and of the corrected statistics under four references.

Reference	SC.HL	SC.EDGE
uncorrected statistic, maximum likelihood covariance	$< 10^{-5}$	$< 10^{-5}$
first order (Ω_{MLE})	0.163	0.128
variance at the debiased fit (Σ_d)	0.024	0.034
de-noised variance (Σ_{dn})	0.007	0.011
CALM	0.012	0.033
prepivotd, 499 refits (Ebrahim, 2026a)	0.026	0.034

estimate does not exist, so the scale estimate operates in a regime the simulations did not probe. And the misfit is in the cubic direction: the three EDGE coordinates of v are -0.16 , 0.14 and -1.38 , so a degree-two basis would return 0.57 rather than 0.033 . A third caution follows from the first. Because the rule keeps the cubic, the EDGE reference here carries the constant 1.15 of Fitted relation 1, and that constant does the work: without it the p -value is 0.011 . This is the one configuration in which the adaptive basis is not free of a fitted constant, and the simulations do not exercise it, since in every cell with $\kappa > 0.05$ the rule keeps the cubic in at most 31 per cent of replications. The reading does not turn on it — every reference in Table 10 but the first places the p -value between 0.007 and 0.034 — but the number should be read as carrying that constant. The computation takes under half a second against 3.3 seconds for prepivoting with 499 refits, and its p -value carries no Monte Carlo error. The conclusion of the earlier paper stands, and no longer depends on a bootstrap.

7. Discussion

Two lessons generalise beyond this test. The first is Theorem 1. Every residual diagnostic for a penalised generalised linear model that standardises by the fitted variance inherits an error of order λ/n in the direction of conservatism, because shrinkage moves fitted values towards the region where the variance function is largest. The grouped tests treated here are the case that clinical prediction practice uses, but the mechanism is not specific to them, and we would expect Pearson-type and deviance-type statistics for penalised fits to show it as well. The second is about grouping rather than about variance. When a diagnostic groups observations on a quantity that depends on the response, the grouping selects on that response, and in the proportional regime the resulting non-centrality does not vanish. We call this response-dependent grouping, and it is not specific to the tests treated here: a calibration plot, a reliability diagram and a decile table of predicted risk are formed in the same way and inherit the same term. At fixed p it is negligible, which is why the classical literature could ignore it; at $\kappa = 0.25$ it is about 6 per cent of the trace of the null covariance on the decile basis, and almost all of it lies in one direction. A statistic that spends every direction of the grouping must then correct for it with something estimated or calibrated, whereas a statistic confined to a few smooth directions can be built so that the affected direction is not among them. That is a design principle for residual diagnostics under penalisation, and it is the reason the two bases in this paper behave so differently. Section 5.6 shows how far the lesson reaches in practice: every calibration check in clinical use — the uncorrected Hosmer–Lemeshow test, the calibration slope, Spiegelhalter’s z , the unweighted sum of squares — rejects almost every correctly specified penalised model, and the one high-dimensional test that does hold its level, that of Janková et al. (2020), answers a different question and is blind along the index where CALM sees. The two are complements, not competitors.

What is proved and what is calibrated should be kept apart. Theorem 1 and Corollary 1 are proved. Proposition 1 rests on the theorems of Bellec (2025) under a Gaussian design and on the consistency of a one-dimensional scale estimate that we assume rather than prove. All of it rests in turn on the first-order law (3), which is proved at fixed dimension and assumed in the proportional regime; an expansion of the grouped residual at $p/n \rightarrow \kappa$ would close that gap and is the largest piece of theory this paper leaves open. Fitted relation 1 is an empirical law with constants fitted on eight cells from two design families; we present the inflation it defines as a theory-motivated correction of order κ whose constant is calibrated and not derived. On the adaptive basis that constant does no work wherever the rule takes degree two, which it did in every cell examined: removing it changes the size there by at most 0.008 , so the level is held without it at every κ examined. Two calibrated quantities do remain on that basis. Corollary 1

is proved for any threshold $\tau \in (0, 1)$, but the value 0.2 was chosen as the smallest at which the size held across the designs of Table 4; and the aspect ratio 0.05 below which the rule keeps the cubic regardless is a choice rather than a calibrated value, with no design examined between $\kappa = 0.01$ and $\kappa = 0.10$. Both are tuning constants with a stated interpretation rather than coefficients fitted to a non-centrality. The order of the selection term follows from the self-influence argument of Section 4.3, but a derivation of its constant from the mean-field theory is open, and it is worth being precise about how far that theory currently reaches. Its prediction of the per-group means of the *raw* residual r is accurate: correlation 0.97 with the observed shape, and the scale to within 27 per cent. Against the per-group means of the *corrected* residual v , which is where the selection term lives once the displacement has been subtracted, the same prediction correlates 0.23. The mean-field theory therefore accounts for the shrinkage displacement, which $\hat{\mu}$ already removes, and not yet for the term that remains.

Five boundaries are stated plainly. The first is the one Section 5.4 draws between CALM-E and CALM-D: CALM-E held or fell below its level in all seventeen cells that the constants never saw, while CALM-D depends on a constant fitted on eight cells and fails where the fitted index has little spread relative to the self-influence, so a correction of the decile statistic that transfers across design families is open. The second is class balance: Section 5.5 shows that at $\kappa = 0.25$ the smooth basis already fails at a prevalence of 0.30 and neither basis holds its level once the prevalence falls to 0.15, while at fixed dimension the decile statistic is unaffected and the smooth one degrades far more slowly, to 0.130 at prevalence 0.08, so the method as it stands is for outcomes that are not strongly unbalanced, and strictly so in the proportional regime. What fails there is the grouping rather than the reference: forming the groups from the predictors alone returns the size to nominal without touching the variance, so the law of Section 4.2 holds inside the failure region too. The third is the regime: the reference is validated for $\kappa \leq 0.4$ and for penalties that shrink towards zero, the theory is for Gaussian designs with robustness beyond them empirical, and bias-reduced estimators in the proportional regime fall outside it altogether (Section 5.9). The fourth is that the penalty λ is treated as fixed, as in Ebrahim (2026a); the selection effect of choosing it by cross-validation on the same data is not accounted for here, nor, so far as we know, by any test in this class. The fifth is power: the power of any grouped test on a noisy index is limited by Proposition 2, so that at $\kappa \geq 0.25$ what CALM offers is validity with power against the departures that survive the index, not power against all.

Three things the closed-form reference buys over the bootstrap are worth collecting, because each is scattered above. Its p-value carries no Monte Carlo error, which matters when the p-value is read rather than compared with a threshold, as in Section 6. It costs one fit rather than 399. And on the decile basis at $\kappa = 0.40$ it holds its level where the bootstrap does not: the bootstrap rejects 0.074 of correct models there in two independent runs of 500 replications (Tables 7 and 9) against CALM-D's 0.035 and 0.036.

What should an analyst do with this? Fit the model, choose the penalty as usual, and run CALM on the adaptive EDGE basis at the penalty chosen, provided the outcome is not strongly unbalanced; that is the statistic we would report, and the decile version alongside it if a reader expects the Hosmer–Lemeshow form. With few events and p/n an appreciable fraction neither statistic is validated (Section 5.5), and the decile version is then the less badly behaved of the two. The reference conditions on that penalty, and the fourth boundary above applies to it: every design in Section 5 fixes λ before the data are generated, so what choosing it by cross-validation on the same data does to the level is not something we have measured. Read a rejection as evidence that the logistic form is wrong along the fitted index, and a non-rejection as saying nothing about whether the shrunk probabilities are calibrated, which under a penalty they are not. The method covers ridge, generalised ridge and any other smooth penalty that shrinks towards zero; it does not cover the lasso, whose penalty is not differentiable, and the elastic net only through its ridge part.

A practical remark closes the paper. Grouping the observations on an index that depends less on the response — the fitted index of a much heavier ridge fit, or a principal component of the design — also removes the selection term, and in simulation gives a valid test without any calibrated constant. We do not recommend it, and the glaucoma data say why: forming the deciles from a fit at five times the penalty re-ranks 38 per cent of the patients and the evidence for misfit disappears ($p = 0.63$). The misfit in that model is a failure of calibration along the deployed index specifically. A test that groups on a different index tests a different hypothesis. CALM corrects the reference and leaves the question alone.

Conflict of interest

None declared.

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Data availability

The GlaucomaM data are distributed in the R package TH.data on the Comprehensive R Archive Network. All code needed to reproduce every number, table and figure in this paper, with random seeds and session information, is deposited in a public reproduction archive at <https://doi.org/10.5281/zenodo.22467285>. The archive exceeds the size limit for supplementary files and is therefore cited rather than uploaded; the online supplementary material is the supplement described below. The archive contains a reference implementation, `calm.gof()`, which runs from the fitted object of any ridge logistic fit and needs no resampling; it is being added to the R package `ebrahim.gof`, and until it appears there the archive version is the one to use; the prepivoting comparator is `shrink.gof()` in version 2.6.0 of that package, on the Comprehensive R Archive Network.

Supplementary material

Section S1 of the online supplementary material reports the behaviour of the correction under penalties other than the ridge, and Section S2 its behaviour under predictor laws other than the Gaussian.

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